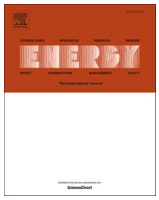
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Review of steam methane reforming as a method of hydrogen production

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ABSTRACT

The article presents a comparison of the performance of the steam methane reforming process in various conditions and scales.

First, a bibliometric analysis of articles related to steam reforming of methane was performed. On the basis of this analysis, a literature review was created in terms of the most popular issues (those with the highest number of reads and the highest number of citations). An important aspect of the selection of topics discussed in the review was that their interest must be new and not historical.

Based on the criteria formulated in this way, the review is divided into non-catalytic and catalytic reforming. As part of the discussion of catalytic reforming, there was also a review of catalysts and their range of applicability (in terms of temperature, fuel composition, pressure, etc.) A further division is made by the scale of the reforming process: laboratory, technical and industrial scale. The article describes methods of meeting the energy needs of the steam reforming process and methods of purifying the hydrogen obtained. The installations for the reforming process presented in the article are summarized in terms of the conditions under which the reaction takes place, type of catalyst used (or none), steam-to-carbon ratio (s/c), installation size, percentage of methane conversion and installation location. As the steam methane reforming is a very broad topic, the article summarizes all the most important issues and shows the forthcoming opportunities of new technologies.

1. Introduction

Hydrogen is the most abundant element in the universe and accounts for as much as 75 % of its mass [1].

Free-state hydrogen on Earth is present in small amounts in the upper atmosphere and in volcanic gases, sometimes in natural gas, as well as in granite and other rocks. Hydrogen in bound form is a component of many organic and inorganic compounds. The greatest amount of hydrogen is found in salt and fresh waters, in hydrocarbons and all organic compounds found in living organisms.

Whereas it is believed that the availability of free-state hydrogen on Earth is very limited, there is a study in which the author has the courage to contradict it [2]. In this comparison, based on various sources, the potential possible annual hydrogen flow in the world from various geological sources was estimated at 23 Tg/year [2]. As this value is an order of magnitude greater than previous estimates, deeper investigation seems advisable.

In recent years, there has been an exponential increase in interest in hydrogen in the context of the energy transformation, which seeks to cut CO_2 emissions. That said, large-scale use of hydrogen predates these concerns.

In 2021 global hydrogen demand was 94 Mt (million tonnes) [3]. A report [3] of the International Energy Agency (Fig. 1) sets out hydrogen demand through 2030.

In 2021, global hydrogen production was predominantly from steam methane reforming (62 %), followed by coal gasification (19 %, mostly in China), naphtha reforming (18 %) in refineries as a by-product, among other methods [3]. Water electrolysis accounted for only 0.1 % of global hydrogen production in 2021 [3].

According to the International Renewable Energy Agency, hydrogen can have the following applications [4]: industrial (refining, production of methane and ammonia, steel production (DRI-direct reduction iron)), energy (flexible power generation, off-grid power, large-scale energy storage), power-to-fuel (renewable gases, synthetic fuels, ammonia), heating (industrial heating, households and commercial heating), transport (road, rail, air, sea transport).

Hydrogen technologies enjoy rich development potential, as evidenced by the report on new projects related to the use of hydrogen [5].

The large-scale use of hydrogen is bound up with multiple regulatory issues [6]. describes the legal aspects related to the use of hydrogen in Australia, Finland, France, Germany, Italy, Singapore, Spain, the

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Abbreviations

aMDEA	activated methyldiethanolamine
APS	announced pledges scenario
CCUS	carbon capture, utilization and storage
CHP	combined heat and power
DRI	direct reduction iron
DME	dimethyl ether
eSMR	electrical steam methane reforming
LHV	lower heating value
LT-PEMFC	low-temperature proton-exchange membrane fuel cells
MAPE	mean absolute percentage error
MCFC	molten carbonate fuel cell
MDEA	methyldiethanolamine
MEA	monoethanolamine
PEMFC	proton exchange membrane fuel cell, polymer electrolyte membrane fuel cell
POX	partial oxidation
PSA	pressure swing adsorption
SMR	steam methane reforming
SR	steam reforming
STEPS	stated policies scenario
THR	thermochemical heat regeneration
USD	United States dollar
WGS	water gas shift
YSZ	yttria-stabilized zirconia

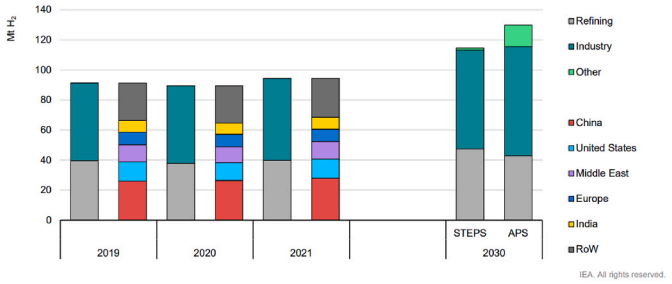


Fig. 1. World demand for hydrogen and forecast demand according to STEPS - Stated Policies Scenario, APS - Announced Pledges Scenario **(source:** [3]**).**

Netherlands, United Kingdom, Portugal and in the European Union generally.

Steam methane reforming (SMR) has been widely studied because of synthesis gas industrial applications and as a primary stage for fuel cell grade hydrogen production. Methane may also be obtained from natural feedstocks such as biomass and biogas. SMR is particularly relevant in emerging green energy technologies [7].

SMR in various formats started to be used for hydrogen production in the early 20th century [8]. By the end of the 1920s, commercial hydrogen production installations were in operation in Baton Rouge, Louisiana and Bayway Refinery, New Jersey [8]. Industrial hydrogen production plants typically use the steam reforming process. Despite its undesirable greenhouse gas emissions, the steam reforming of natural gas is the most common method for hydrogen production. This method is also known as SMR because methane is the primary component of the natural gas. This method currently meets approximately fifty percent of global hydrogen demand. The other fifty percent comes from oil and naphtha reforming in refinery (around thirty percent) and chemical industrial off gases; eighteen percent comes from coal gasification; three point nine percent comes from water electrolysis; and one point one percent comes from other sources [9].

Numerical studies also play a very important role in the steam reforming optimization process. In Ref. [10] a new design of autothermal reformer was presented and the influence of catalytic layer patterns was investigated. This resulted in shortening the reactor length by 48 % compared to conventional autothermal reactors. For the purpose of optimizing the SMR process, CFD (Computational Fluid Dynamics) methods are also useful. In Ref. [11], the influence of different types of catalyst coating (continuous and discrete catalytic layer) was investigated using CFD methods. Using CFD methods in Ref. [12], multiobjective optimization of membrane SMR reactor was performed. High hydrogen separation was achieved while reducing the amount of catalyst by 20 % and shortening the membrane length by 80 % compared to a conventional reactor. In Ref. [13], a microkinetic model was used to simulate the operation of a plate reactor with segmented and continuously coated layers of catalysts. Methane steam reforming and methane combustion were investigated. To achieve similar methane conversion and hydrogen yield, compared to the reference case, the amount of combustion catalyst needed was 66 % less using the segmented catalysts design.

Steam methane reforming is a very broad topic, this article highlights key issues and explores emerging technological opportunities. The article reviews the state of knowledge about the SMR method of obtaining hydrogen, from the point of view of scale and the conditions in which this process is carried out. Moreover, the article describes in detail the energy management of steam methane reforming processes, which is not included in the previously published review articles.

2. Bibliometric analysis

The bibliometric analysis was carried out based on the results obtained from the Scopus and Web of Science databases. This analysis was performed on December 31, 2024. These databases were searched for articles about steam methane reforming.

The first step was to check whether the topic of steam reforming of methane is still relevant. For this purpose, the number of scientific articles on this topic published in recent years was determined and the results are shown in Fig. 2.

The number of scientific articles published every year on this topic is constantly growing, which allows us to assume that the topic is attractive to scientists.

It was also checked which authors published the most on steam reforming of methane, as shown in Fig. 3.

In the next step, the Scopus database was asked about the affiliation of the authors publishing the most articles (in years 2020–2024) on this topic and the results are shown in Fig. 4.

This question was also repeated in the context of the country in

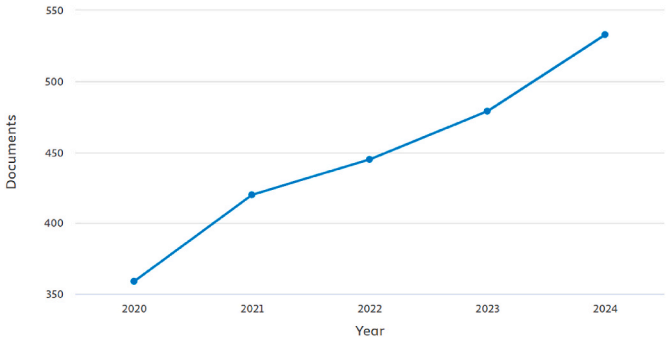


Fig. 2. Number of scientific articles on steam methane reforming published in 2020–2024 (source: Scopus database).

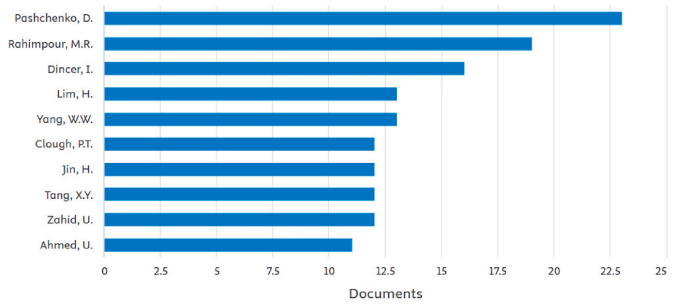


Fig. 3. Number of scientific articles on steam methane reforming published in 2020–2024 written by the 10 authors with the most articles on this topic (source: Scopus database).

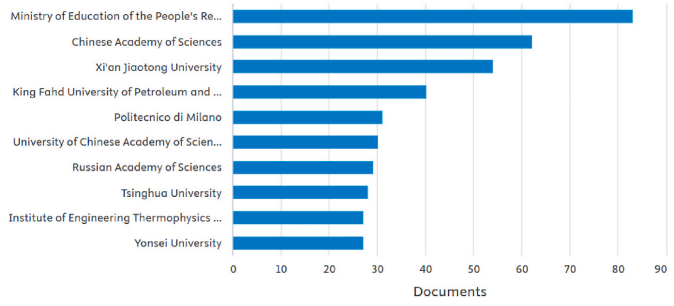


Fig. 4. Number of scientific articles on steam methane reforming published in 2020–2024 with the affiliation of research centers with the most articles on this topic (source: Scopus database).

which the authors publishing the most articles on steam reforming of methane are affiliated (Fig. 5).

It turns out that this topic is very popular in China, while in other countries of the world it is less popular, which is (in the authors' opinion) a direct result of the climate policy of Europe and the United States. Another possibility for these results is that China has a larger population than either Europe or the United States.

The purpose of this paper is to write a review article on steam methane reforming that will be successful. At this point, we need to answer the question what parameter can mean that a review article can be considered a success? In the authors' opinion, such a parameter should be the number of citations of article on a given subtopic of a selected research area.

For this purpose, the citation numbers of articles on steam reforming of methane divided into subtopics of the article were analyzed in the Web of Science database in the years 2020–2024 (see Fig. 6).

It turned out that the subtopic related to catalysts for the reforming process was the most popular, so an entire section of this article was devoted to this topic.

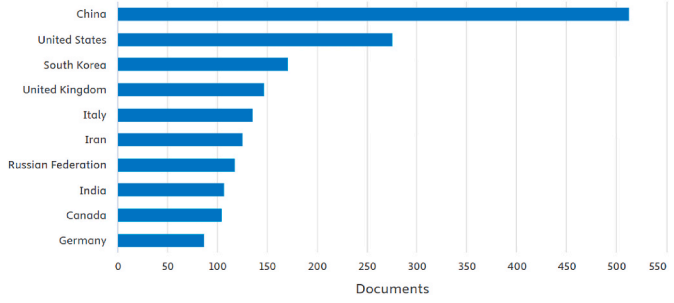


Fig. 5. Number of scientific articles on steam methane reforming published in 2020–2024 - by country (source: Scopus database).

We also checked in the Web of Science database which research areas enjoyed the highest citations (Fig. 7) and the results of these searches indicate: Energy Fuels, Engineering Chemical and Chemistry Physical.

Therefore, this review article has been written to also take these criteria into account.

3. Mathematical description

Mathematical models of steam reforming processes usually require involving reaction kinetics. Kinetic models elucidate the influence of many factors, such as concentration, pressure, and temperature, on the speeds of chemical processes. These models assist in forecasting the behavior of reactions under various conditions, hence facilitating the design and optimization of reactors and processes. The fundamental nature of a kinetic model resides in its capacity to quantitatively articulate the connections between response rates and the variables that affect them. Usually, these models consists of equations that include parameters such as reaction rate constants, which may change with temperature based on the Arrhenius equation, and reaction orders, which show how the reaction rate is influenced by the concentration of each reactant. Often kinetic models are very complex and are based on the experimental data.

Most reliable models that included catalytic methane reforming kinetics were presented by Fu et al. [14] who defined reforming process through two parallel reactions:

$$CH_4 + H_2O \leftrightarrow CO + 3H_2 \ [\Delta H = +210\ \text{kJ/mol}] \quad (1)$$

$$CH_4 + 2H_2O \leftrightarrow CO_2 + 4H_2 \ [\Delta H = +160\ \text{kJ/mol}] \quad (2)$$

The researchers conducted their studies within a temperature range of 400 °C–450 °C, using a nickel-based catalyst infused with rare earth metals with S/C ratio 2:1. Total reaction rate equation based on the experimental results:

$$r_{CH_4} = r_{CO} + r_{CO2} = k_{CO} p_{CH_4}^{0.8} + k_{CO2} p_{CH_4}^{0.8} p_{H_2O}^{1.5} \quad (3)$$

Another well presented results, which allows to believe that the proposed model is reliable were published by Souentie et al. [15], who conducted experimental study on methane steam reforming over Ni/GDC and AueNi/GDC, a fuel cell anode material at higher temperatures ranging from 800 °C to 900 °C. The steam to carbon ratios used ranged from 0 to 1.25. The reaction rate of CH_4 conversion was defined by equation:

$$r_{CH_4} = \frac{k p_{CH_4}}{1 + K \frac{p_{CH_4}}{p_{H_2O}}} \quad (4)$$

The reaction rate relationship was also proposed by Sciazko et al. [16]:

$$r = k p_{CH_4}^{0.88} p_{H_2O}^{-0.83} \quad (5)$$

This formula is valid for reaction temperatures in the range 550–750 °C and S/C ratios from 3:1 to 6:1.

A slightly different equation describing the reforming reaction is presented in Ref. [17]. This formula is applicable to the temperature range from 500C to 650 °C and is showed below:

$$r = k p_{CH_4} \quad (6)$$

A more detailed description of the methane steam reforming process is presented in Ref. [18]. The steam-to-carbon ratio ranged from 3:1 to 5:1, with temperatures ranging from 773 K to 848 K. The authors assume three reactions occur at the same time: methane steam reforming with CO as a product, MSR with CO_2 as a product, and the water-gas shift reaction.

CO as product:

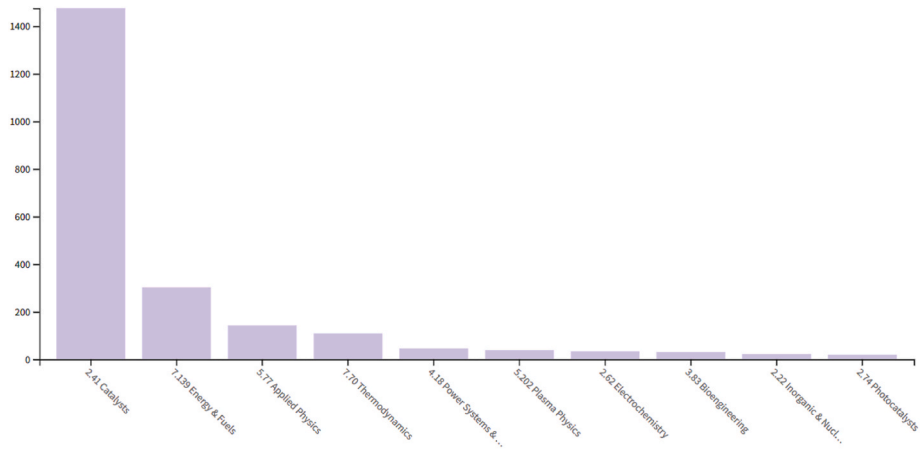


Fig. 6. Citations of articles collected in the years 2020–2024 divided into subtopics of the article related to steam methane reforming (source: Web of Science database).

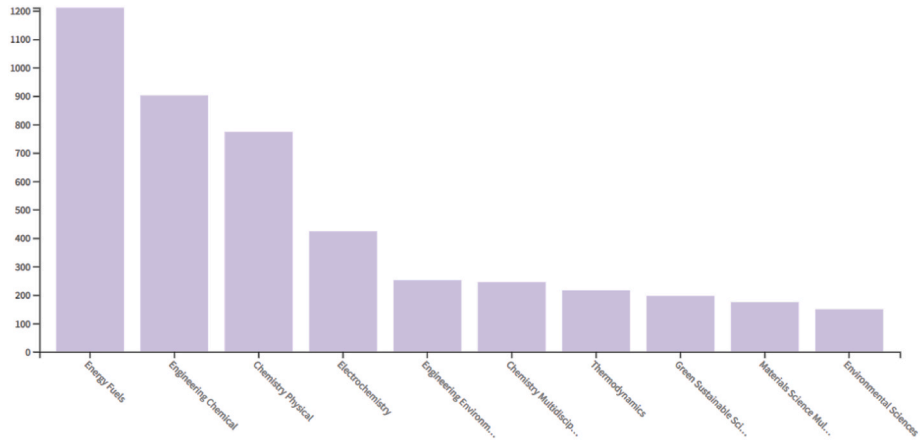


Fig. 7. Number of articles related to steam methane reforming published in 2020–2024 - by research area (source: Web of Science database).

$$r_1 = \frac{\frac{k_1}{p_{H_2}^{2.5}}\left(p_{CH_4}p_{H_2O} - \frac{p_{H_2}^3 p_{CO}}{K_1}\right)}{DEN^2} \tag{7}$$

WGS reaction:

$$r_2 = \frac{\frac{k_2}{p_{H_2}}\left(p_{CO}p_{H_2O} - \frac{p_{H_2}p_{CO2}}{K_2}\right)}{DEN^2} \tag{8}$$

CO_2 as product:

$$r_3 = \frac{\frac{k_3}{p_{H_2}^{3.5}}\left(p_{CH_4}p_{H_2O}^2 - \frac{p_{H_2}^4 p_{CO2}}{K_3}\right)}{DEN^2} \tag{9}$$

where: $DEN = 1 + K_{CO}p_{CO} + K_{H_2}p_{H_2} + K_{CH_4}p_{CH_4} + K_{H_2O}\frac{p_{H_2O}}{p_{H_2}}$.

The mathematical model according to formulas (7)-(9) was used in Ref. [19] to describe steam reforming of methane in packed bed reactor with internal helical heat fins.

More on the mathematical relationships used to describe the methane steam reforming process (with particular emphasis on reforming for the needs of fuel cells) can be found in the authors' previous work [20].

4. Non-catalytic reforming

Non-catalytic reforming takes place at a much higher temperature than prevails in the presence of a catalyst. According to Ref. [21] at above 1000 °C methane decomposes into radicals that can create ethylene, acetylene and coke. At temperatures above 1500 °C these radicals can then react with steam radicals [22].

The research on non-catalytic steam reforming available in the literature only concerns modeling [23] and theoretical considerations [21,23].

In [23] the kinetics of the reaction were modeled and the process of non-catalytic steam reforming of methane and higher hydrocarbons (up to C4) at temperatures of 1126.85–1526.85 °C was thermodynamically analyzed. The sequence of hydrocarbon transformations and the duration of these transformations were determined. The first stage of conversion is pyrolysis to acetylene and hydrogen without steam. The conversions involving water vapor begin much later. OH radicals are formed from water vapor and then react with acetylene, resulting in the formation of carbon monoxide and hydrogen.

The literature includes works on steam reforming of non-methane fuels such as bio-oil [24,25], ethanol [26], and C2-C4 hydrocarbons [23]. Experimental studies have been conducted for partial oxidation of methane [27–29] and other fuels [27,30]. Internal reforming process taking place in high temperature fuel cells has been also identified in alcohols [31,32]. Pedersen-Mjaanes et al. used a two section, non-catalytic fuel reformer similar to those studied by Kennedy's group to extract hydrogen from various fuels [27]. The literature also contains articles on the modeling of partial oxidation of methane [29,33–39] and other fuels [39].

5. Catalytic reforming

The steam methane reforming is process in which methane reacts with steam to produce a mixture of hydrogen, carbon dioxide and carbon monoxide. In order to obtain acceptable reaction rates a catalyst is required to accelerate the process. The catalyst should be stable under the rate extreme conditions of reactions for example high temperature and for unwanted side reactions involving carbon deposition. Carbon deposition is one of the most important issues in SMR reactors, being dependent on the temperature, pressure, catalyst and S/C ratio [40]. The relation of residence time of methane in the reactor and time is linear, thus catalysts bed requires regeneration. Generally, Ni (the base metal) or the noble metals Rh, Ru, Pd, Pt, Ir are used as the active metals in catalysts for steam methane reforming. Nickel is a widely used transition metal because due to its low cost, a good performance and easy availability. Disadvantage of Ni based catalyst is that it is a less active than metals from this set and more prone to sintering and coke formation. A comprehensive review of research on Ni based catalysts was presented by Jang et al. [41]. In search of catalysts with better performance, researchers have been added a various elements as promoters to conventional Ni-based catalysts in order to improve its selectivity, durability and activity [42–45]. The noble metals are most commonly used as a promoter for Ni-based catalysts. They are characterized by their high catalytic ability, stability in high temperature applications and resistance to carbon deposition. Even though, they are considered uneconomical in industrial applications due to their high costs. A pioneering study on the activity of these metals done by Kikuchi et al. [43] suggested the order Rh ~ Ru > Ni > Ir > Pd ~ Pt ≫ Co, Fe. Later research by Rostrup Nielsen and Hansen observed relative activities in the following order: Ru ~ Rh > Ir > Pt ~ Pd [44]. Qin and Lapszewicz [45] suggested the activity relationship for noble metal catalysts in the SMR reaction as Ru > Rh > Ir > Pt > Pd. Jones showed the activity order of metals in a different way than others authors as Ru > Rh > Ir > Pt [46]. In study [47], Wei and Iglesia showed that supported Ir and Pt catalysts are more active than Ru and Rh at high temperature (873 K). Morales-Cano et al. [48] presented the use of noble metals Pt, Pd, Rh, Ru, Ir as promoters for Ni-based catalysts (Ni/α-Al_2O_3) for SMR. Nieva et al. [42] recommended that the catalytic activity of Ni-based catalysts mainly depends on the support, which plays an important role in the catalytic process. He suggested the activity order of Ni-based catalysts with different supports as Ni/$MgAl_2O_4$ > Ni/$ZnAl_2O_4$ > Ni/Al_2O_3 > Ni/SiO_2.

Review of metals employed as catalysts for SMR in shown in Table 1.

Many researchers reported the effect of synthesis methods of catalyst addition to SMR with modified supports such as Al_2O_3, MgO, CeO_2, TiO_2 and promoters (using Ce, Co, Cu, Fe, Pd, Pt, Rh, Ru, Zr) to obtain high yields. Tada et al. [117] confirmed use a catalyst support of $Ce_2Zr_2O_7$ on ni-based catalyst in SMR. Sutthiumporn et al. [118] showed the effect of adding non-noble metals such as Bi, Co, Cr, Cu, Fe in an $La_{0.8}Sr_{0.2}NiO_3$ perovskite structure as a precursor for a dry reforming methane catalyst. The effect of Ag content on the catalytic properties of Ni-based catalyst was by Lazar et al. [119–121] and Bueno et al. [122]. Whereas, Maniecki et al. [123]. recommended the addition of group 11 metal (Au) to Ni/Al_2O_3 catalyst. Others as Shao et al. [124] have been interested the addition of lanthanide promoters such as La_2O_3, CeO_2, Pr_2O_3, Gd_2O_3, Sm_2O_3 to Ni/Al_2O_3 catalyst. Kim et al. [125] studied the effect of Ni/$MgAl_4$, Ni/Al_2O_3, Ni/CeO_2, and Ni/$Ce_{0.4}Zr_{0.6}O_2$ catalyst for SMR. Others as Roh et al. [126] tested performances of Ni-based catalysts as Ni/Ce-ZrO2/θAl2O3, Ni/θ-Al2O3, and Ni/γ -Al2O3 in research of SMR. However, Zhai et al. [127] studied performances of Ni/ZrO_2/Al_2O_3, Ni/LaCa/Al_2O_3, and $Ni_{0.5}Mg_{2.5}AlO_9$ catalysts for SMR. Other research was done by Bej et al. [128]. Salhi et al. [129] researched the effect of wt.% NiO in $NiAl_2O_4$ catalyst for SMR process.

Examples of various catalysts for steam reforming of methane are shown in Table 2.

Various types of metals can be catalysts for the SMR process - see Table 3.

Steam reforming of other fuels may require the use of different catalysts. For example, copper-based catalysts can be used in the methanol steam reforming process [152]. Another interesting issue is diffusion limitation in the SMR process which refers to the restriction on mass transfer within the catalyst particles, affecting the efficiency of the reforming reactions. In the context of SMR, diffusion limitations arise primarily due to the size and porosity of the catalyst particles. Larger catalyst particles tend to have significant internal diffusion resistance, reducing the accessibility of reactants to active sites within the particle's core. This leads to a lower effectiveness factor, which is a measure of the actual reaction rate compared to the ideal rate without diffusion constraints. The effectiveness factor decreases with increasing particle diameter and is inversely related to the temperature and steam-to-methane ratio, indicating more pronounced diffusion limitations under these conditions [153].

6. Laboratory scale reforming

SMR is the most widely used process for hydrogen production. The hydrogen obtained from this method is used in industrial processes and in fuel cells because of its purity. Many authors have presented experimental results of SMR at the laboratory scale. In Ref. [154] the author presented experimental results of SMR performed over an Ni based catalyst at relevant conditions. The kinetics of the SMR reaction in laboratory scale fixed-bed catalytic tubular reactors was the subject of a study by Costamagna et al. [155]. Ściążko et al. [156] presented experimental (Ni/YSZ catalyst) and numerical results on SMR kinetics using their selected numerical method. The effect of operating parameters such as various temperatures and space velocities on SMR was

Table 1
– Various metals in SMR.

Noble metals		Non-noble metals	
Ru	[49–54]	Fe	[55–67]
Rh	[48,68–71]	Co	[72–81]
Re	[82–84]	Cu	[60,85–90]
Ag	[91,92]	Sn	[93–96]
Ir	[48]	Mn	[97,98]
Pt	[99–103]	Mo	[104–106]
Au	[107–112]	In	[113,114]
		Zn	[115,116]

Table 2
Summary of example catalysts for SMR.

Catalysts			
Monometallic	Ref.	Polymetallic	Ref.
Ni/$SiO_2Al_2O_3$	[130]	xNi-yCu-tZn/γ-Al_2O_3	[131]
Ni/$MgAl_2O_4$	[132]	xNi-yLa/α-Al_2O_3	[133]
$NiAl_4O_7$	[134]	Ni/$MgAl_2O_4$	[135]
$Ni_2Al_2O_5$		$NiMgAlO_x$	
		Ni/Mg/γ-Al_2O_3	
Ni/$Y_2Zr_2O_7$_x	[136]	xPt/15 wt%Ni/$MgAl_2O_4$	[99]
x = HT, CP, GNC			
Ni-$La_{0.6}Sr_{0.6}FeO_{3-\delta}$	[61]	Ni/$CaZrO_3$/α-Al_2O_3	[137]
Ni-$SrTi_{0.7}Fe_{0.3}O_{3-\delta}$			
Ni/$Ce_{0.95}M_{0.05}O_{2-d}$	[138]	Ni-X/α-Al_2O_3	[48]
Ni-γ-Al_2O_3	[139]	LA-Ni/ZrO_2	[140]
Ni/$Ni_{0.4}Mg_{0.6}O$	[141]	Co-M_1-M_5/Al_2O_3	[142]
Ni/$Y_2B_2O_7$	[143]	Pt,Pt,Ni/$Mg_{1-x}La^{3+}$	[144]
mNi/xLa-Si	[145]	–	–
Ni/γ-Al_2O_3	[146]	–	–
Ni/$Ln_2Ti_2O_7$	[147]	–	–
$NiAl_2O_4$	[148]	–	–
Ni/Al_2O_3	[149]	–	–
Ni/$ZnLaAlO_4$	[150]	–	–
Pt/$ZnLaAlO_4$			
Ru/$ZnLaAlO_4$			

Table 3
Example metals as catalysts of SMR.

Metal						
Monometallic		Bimetalic				
Non-noble	Noble	Non-noble	Noble	Mixed		
Ni/ZrO_2	Rh/Ta_2O_5 or ZrO_2	$Ni-ZrO_2/Al_2O_3$	–	$Ni-Rh/ZrO_2$	**Reducible metal Oxides**	Support
Ni/Al_2O_3	Rh/Al_2O_3	Ni-Co $Ni-Ce/Al_2O_3$	Pt-Ru/γ- Al_2O_3	$Ni-Rh/Al_2O_3$	**Irreducible metal oxides**	
Co/Activated carbon	–	Ni-Co/SBA15	–	Rh-Ni/SBA15	**Mesoporous**	
Ni/MgO- Al_2O_3	Rh or Perovskites	$Ni-Mo/SBA15-La_2O_3$	–	–	**Composites**	

An overview of nickel-based catalysts for the methane steam reforming process is presented in Ref. [151].

studied. Experiments were performed in bench scale reactors [157] (Fig. 8).

Laboratory scale reforming of methane presented in article [158] which showed a test of the effects of a commercial catalyst and a laboratory-made catalyst on SMR. In Ref. [159] the authors tested for the effects of catalysts of various Ni percentages on SMR in a fixed bed reactor. Catalytic activity and selectivity for various Ni content Ni/Ce-ZrO2 catalysts were tested in Ref. [160]. CO_2 reforming of methane into syngas over an $Ni/CaO-Al_2O_3$ catalyst was studied by Lemonidou et al. [161]. Oliveira created a model with diffusion to study the kinetics of SMR at different operating conditions and for a selected catalyst [162]. SMR was studied in Ref. [163] on stable Ni on an alumina-titania based catalyst.

Research on an MCFC fuel cell fueled with methane supplied by steam reforming taking place in the anode channel containing a catalyst was placed is presented in Ref. [164]. These tests were carried out on the test stand shown in Fig. 9. During the tests, a series of current-voltage curves were determined for operating temperatures of the fuel cell of 550 °C and 650 °C and s/c ratios of 2, 2.5, 3 and 3.5. In case of fuel cells the interesting tests are also a comparison of the same fuel cell working with methane fuel and liquid fuels such as methanol, ethanol etc. [32].

Garbarino et al. [158] tested commercial and laboratory-made Ni-based catalysts for CO_2 methanation and SMR for several operating conditions. Other authors [159] studied activities of catalysts of varying Ni content in a fixed bed reactor for SMR. Oliveira et al. [162] determined the kinetics of SMR using a large pore Ni/Al_2O_3 catalyst. The kinetic parameters of SMR over an $NiO/\alpha-Al_2O_3$ catalyst were determined by comparing the results of a mathematical model of the SMR process with experimental results [165]. A diagram of their laboratory stand is shown in Fig. 10.

Experimental research [166] on steam reforming of fuels such as methane, biogas and dimethyl ether (DME) was conducted at the Institute of Power Engineering in Poland.

The reformer (Fig. 11) was built in the form of a plate exchanger with corrugated plates covered with a catalyst (on the side of the heated medium). The tests were carried out for a temperature range 500–750 °C and for s/c coefficients of 2, 2.5, 3 and 3.5.

In the next step, the Warsaw University of Technology built a mathematical model representing the behavior of the reformer. The calculated error of the reformer model did not exceed 3.5 %. The results of the experimental research and modeling were published in a joint article [166]. The mean absolute percentage error was used to measure the model error in relation to the measured values according to the following relationship:

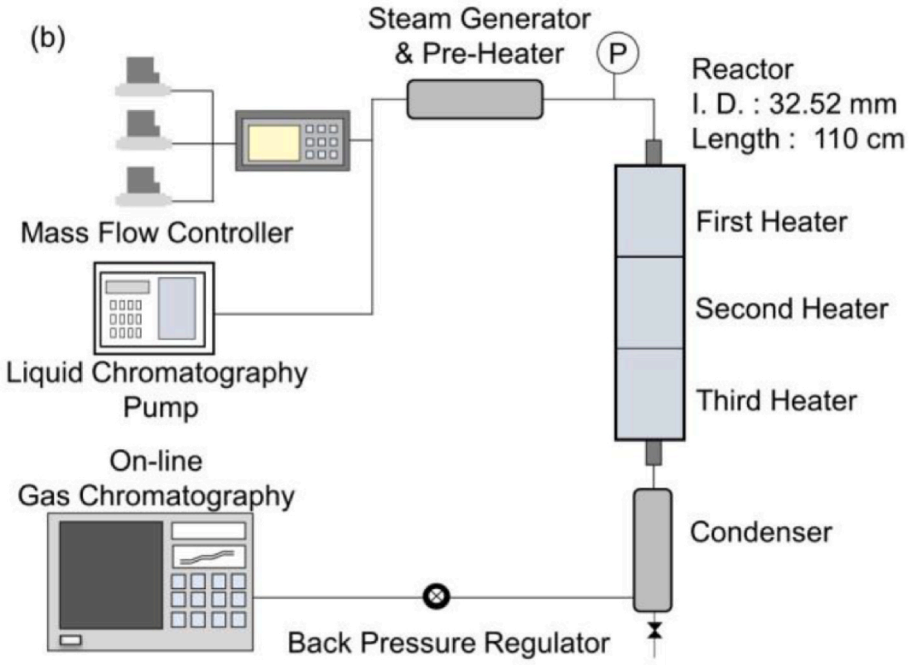


Fig. 8. Schematic of the SMR system in Ref. [157].



Fig. 9. Laboratory stand for testing high-temperature fuel cells owned by the Institute of Heat Engineering of Warsaw University of Technology.

$$MAPE = \frac{100\%}{n} \sum_{t=1}^{n} \left| \frac{A_t - F_t}{A_t} \right| \tag{10}$$

where: A_t - measured value, F_t - value obtained from the mathematical model, t - number of measured or calculated value, n - total number of measured or calculated values. The values generated by the mathematical model were the volumetric composition of the gas produced as a result of the reforming process. An example comparison of the measurement data and the results obtained from the model is shown in Fig. 12.

This figure shows the results of the comparison for methane after calibration of the model.

Interesting case is also combined reforming investigations, which are focused on the application of other gases except pure water (such as flu gas from combustion products or other H2O-CO2 mixtures). Those

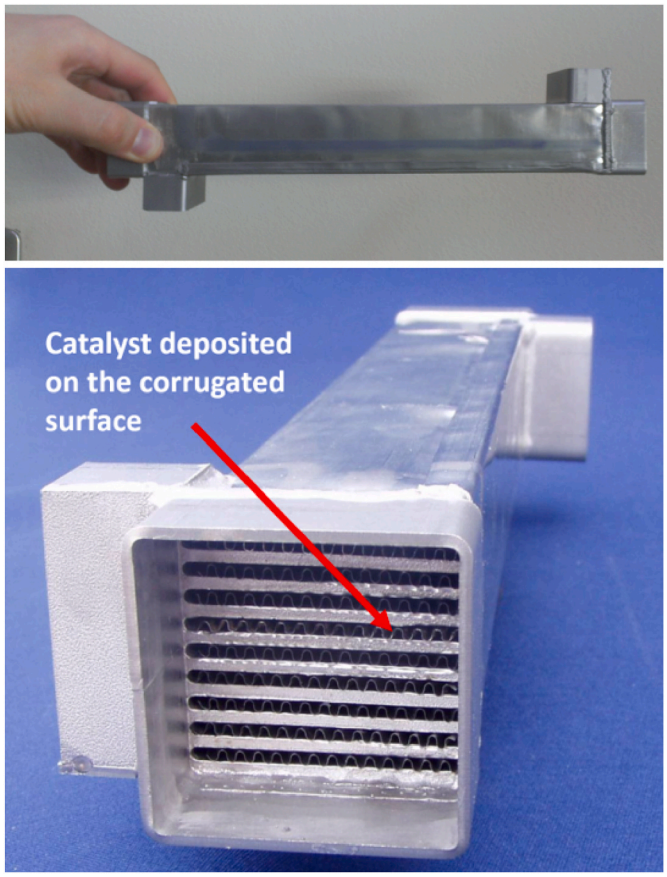


Fig. 11. Reformer belonging to the Institute of Power Engineering [166].

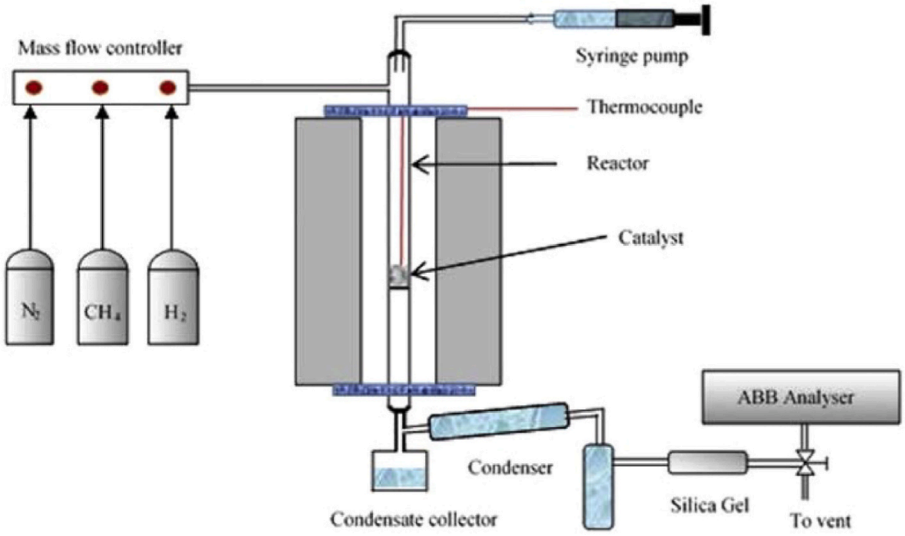


Fig. 10. Schematic of apparatus described in Ref. [165].

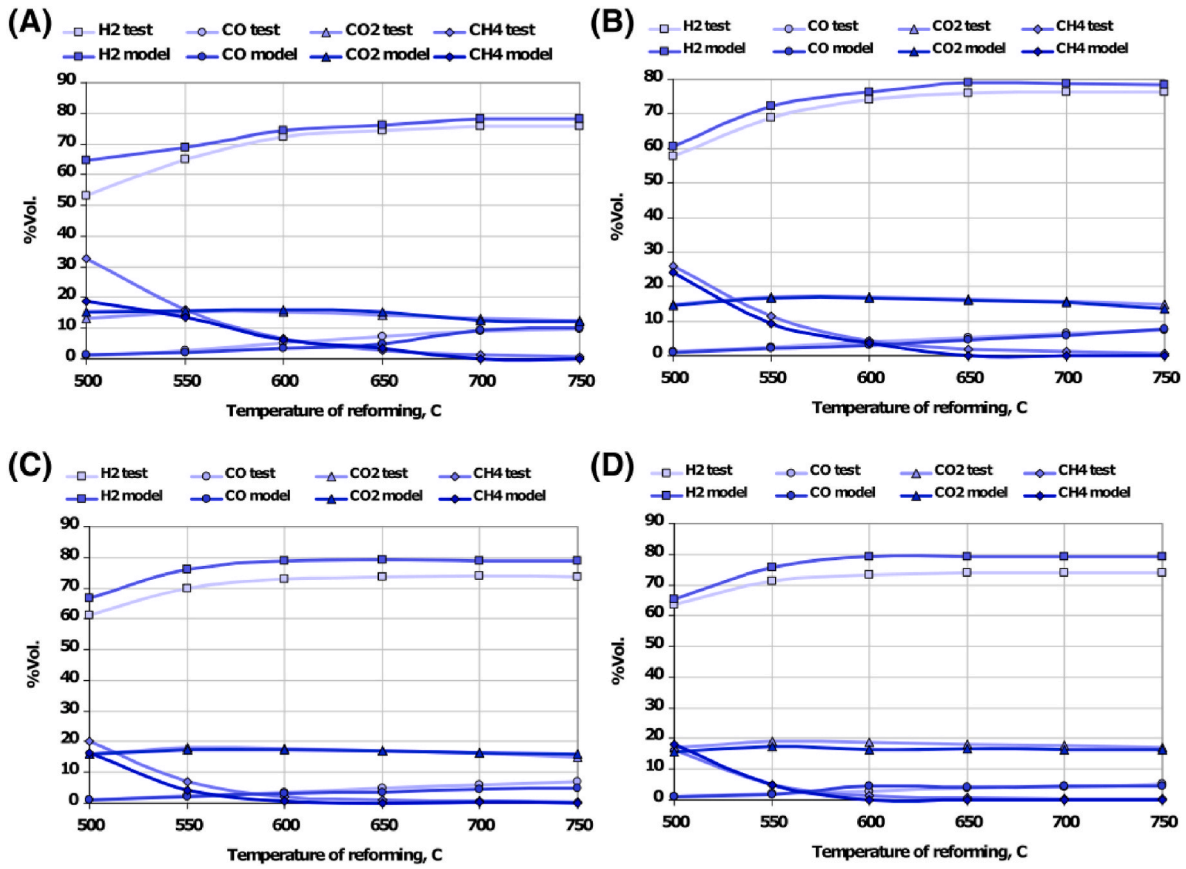


Fig. 12. Comparison of the volumetric compositions of gas after SMR (for s/c = 2, 2.5, 3 and 3.5) with the results of the mathematical model **(source:** [166]**).**

solutions can be perspective ways for increasing CO2 utilization and promote circular economy [167,168].

Another interesting idea is electrified steam methane reforming [169,170]. Electrified methane steam reforming incorporates direct electrical heating into the process, utilizing ohmic heating for uniform temperature distribution and quick thermal response. This approach minimizes carbon buildup and thermal gradients, improving efficiency and control. It offers flexible operation and can reduce CO2 emissions by using renewable electricity for heating, enhancing sustainability compared to traditional gas-fired methods.

7. Reforming on a technical and industrial scale

Steam reforming of methane or natural gas is currently the most common method for hydrogen production worldwide. Technical and industrial scale SMR plants usually have two purposes: to be as cheap as possible (in the context of their objective) and to satisfy the required volume of production (usually large). There are a few indicators for technical and industrial scale SMR facilities, such as minimum unit price of production (price/kg) and optimum production volume (kg/year). SMR meets usually both of these indicators as well as investment cost: this technology can be scaled up simply.

In 2020, the global hydrogen demand of 90 Mt was satisfied nearly exclusively by hydrogen derived from fossil fuels, with 72 Mt H_2 (79 %) coming from specialized hydrogen production units (see Fig. 13). 72 Mt H_2 was required for pure hydrogen, mostly for ammonia production and oil refining, whereas 18 Mt H_2 was combined with other gases and used for methanol production and DRI steel manufacture [171].

There are four commercial methods for producing hydrogen: catalytic partial oxidation, catalytic partial breakdown of methane, catalytic steam reforming, and noncatalytic partial oxidation. Among them steam reforming is the cheapest and the most popular [172]. The levelized cost

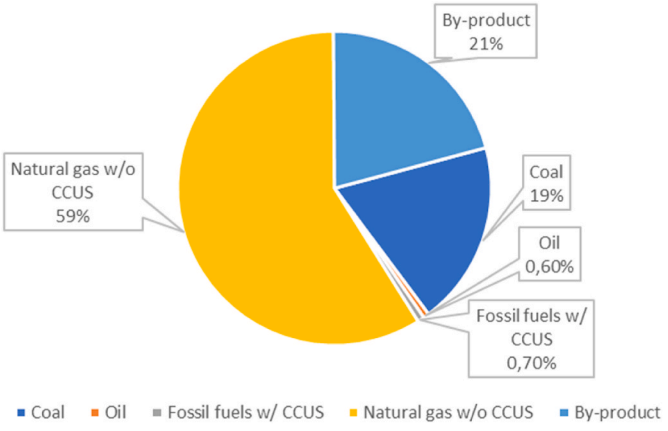


Fig. 13. Global hydrogen supply in 2020 [171] **(CCUS = carbon capture, utilization and storage).**

of hydrogen generated from natural gas ranges from $0.50 to $1.70 per kilogram of H_2, depending on local gas costs. It is substantially more expensive to use renewable energy in most regions, costing USD 3–8/kg H_2. Water is also needed for hydrogen generation in addition to energy. The least water-intensive method is water electrolysis, which uses just 9 kg of water for every kilogram of hydrogen produced. Water usage for natural gas production with CCUS increases to 13–18 kg H_2O/kg H_2, but water use for coal gasification increases to 40–85 kg H_2O/kg H_2, depending on the amount of water used for coal mining. Recent innovations in hydrogen production via natural gas steam reforming focus on CCUS units, as is presented by Shell for example [173]. Commercial operations have begun in Alberta, Canada, as part of the "Quest" carbon

capture and storage (CCS) project. More than a million tons of CO_2 per year will be captured and stored securely by Quest. Investigations on pilot solutions for co-electrolysis (hybrid system for CO2/H2 mixture production) have been also presented by Refs. [174,175].

Steam reforming is usually used for the production of hydrogen for the purposes of deriving ammonia and methanol, as product purity is rather low – usually used directly in the form of synthesis gas ($CO + H_2$). Where pure hydrogen is required, the plant needs to be equipped for CO shifting by the water-gas shift reaction (WGS), aimed at removing the CO_2 present in the gas mixture. Reactor temperature, operating pressure, make-up of the input gas, and the amount of accompanying steam all affect the composition of the reformer's output.

Reformer systems used in industrial settings typically generate between 7 and 30 million normal cubic meters (Nm^3) of hydrogen gas per day. The systems can function at temperatures as high as 850 °C and pressures as high as 2500 kPa (see Fig. 14). They are made up of tubular reactors crammed with catalyst pellets. In general, the reactors are usually about 12 m long and are constructed out of costly alloy steels so that they can withstand the high temperatures and reducing gas conditions. These reformers can be scaled down in a relatively simple manner, giving H_2 outputs of around 0.1–0.3 million Nm^3 per day.

The conventional steam reforming process consists of two reformers (or two separate beds). In the first reformer water (steam) is mixed with methane with steam-to-carbon ratios of between 2:1 and 6:1, usually in higher pressures (5–35 bar). The reaction that takes place in the first reformer is as follows:

$$CH_4 + H_2O \rightleftarrows CO + 3H_2 \ [\Delta H = +206 \text{ kJ / mol}] \quad (11)$$

The first reaction is endothermic [177] thus heat needs to be delivered, usually through the combustion of natural gas. The heat required for the energy balance is up to a few % of the methane used for steam reforming. The CO generated then reacts with the steam (often an additional amount) in the next reactor or bed of the reforming unit – the water gas shift reactor, according to the equation:

$$CO + H_2O \rightleftarrows CO_2 + H_2 \left[\Delta H = -41 \frac{\text{kJ}}{\text{mol}}\right] \quad (12)$$

Absorption of amines is one method that may be used to purify hydrogen, pressure-swing adsorption technology is another. Amine absorption is used in some of the older facilities, while many of the more recent hydrogen production plants prefer adsorption by porous materials [178]. A typical modern steam reforming process scheme is presented below (Fig. 15).

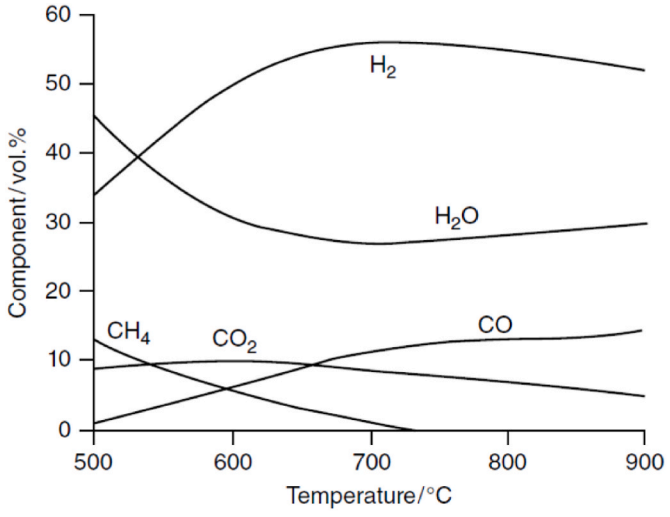


Fig. 14. Equilibrium concentrations of products and reactants for SMR at 100 kPa as a function of temperature [176].

Catalysts applied in steam reformers are usually Ni-based, possibly with other additives. Since Ni is the best price/performance catalyst, it is usually used in commercial applications. Numerous valuable metals including platinum, palladium, and rhodium perform better but are much more expensive. The SR catalyst should ensure coking resistivity or decomposition by steam, high activity at sustained elevated temperatures, activation only for desired reactions and resistance to mechanical stresses (such as gas and steam flows, high pressures etc.). As the reforming reaction occurs on the surface of the catalyst, the optimal shape is one with a high area – often the catalyst is produced in the shape of porous materials (such as net or foam), but the most resistant, preferred options are perforated cylinders or cubes made of ceramic, covered by an Ni layer.

In the case of the catalysts produced by Ref. [179] the catalysts G-0117, G-0117-7H, G 0117 H/C, G-0117K-7H/C, and G 0217 7H/C have Ni placed on a porous ceramic support as its active component (see Table 4). The carrier, produced at extremely high temperatures, imparts excellent activity, thermostability, and mechanical strength to the catalysts. The catalysts are provided as oxides. To obtain the active state, NiO must be reduced to Ni in an industrial reformer after being loaded. The G-0117K-7H/C catalyst is further promoted with potassium, while the G 0217-7H/C catalyst is constructed on a $CaO\text{-}Al_2O_3$-type support, which, under particular process circumstances, increases their pendency resistance. The catalysts are shaped as rings (G-0117), seven-hole cylinders with flat or spun bases (G-0117-7H/C, G-0117K-7H/C), and seven-hole cylinders with gouges and spun bases (G-0217-7H/C).

Unfortunately, other commercial catalyst producers, such as Clariant [180], Umicore [181] and Johnsson Matthey [182] do not publish information about the precise composition of their catalyst or functional parameters. Tanaka Precious Metals [183] for example deliver very detailed parameters and characteristics of performance of their catalysts, but they are based on rare earth metals with very high prices.

In addition to the catalyst applied, reactor design has an impact on the efficiency and possible yield of hydrogen, which are significant factors in large scale industrial applications.

For example [184] presented a complex optimization of methane steam reformer, considering temperature distribution, gas mixture velocity and catalyst distribution in the reforming reactor. The results show that a dense catalyst package is not always desired, as it often implies local temperature drops, decreasing possible hydrogen yield. The fact that reactions (1) and (2) are either endothermic or exothermic has a large impact on reactor construction. However, the optimal design and catalyst bed load can be determined for various shapes and lengths of reactor and cannot be improved over this optimum. In the case of catalysts, research to find better and cheaper catalysts materials is still ongoing.

Some prospective, cheap mixtures based on Ni were analyzed in Ref. [125]. The researchers studied the impact of the support for the Ni catalyst, which included $Ni/MgAl_4$, Ni/Al_2O_3, Ni/CeO_2, and $Ni/Ce_{0.4}Zr_{0.6}O_2$ for SMR. Experiments were carried out on supports with an Ni loading of 10 percent at 750° Celsius and a steam-to-carbon ratio of 2. Ni/Al_2O_3 demonstrated CH_4 conversion of 97 percent and a composition of 73.5 % hydrogen, whilst $Ni/MgAl_2O_4$ delivered CH_4 conversion of 97.2 percent and a hydrogen content of 71.5 %. CH_4 conversion; and H_2 composition was found to be lower in Ni/CeO_2 and $Ni/Ce_{0.4}Zr_{0.6}O_2$ than in Ni/Al_2O_3. Another study with cheap additives

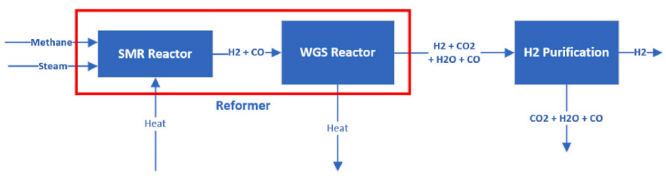


Fig. 15. Simplified diagram of typical process of industrial SMR for hydrogen production.

Table 4
Commercial catalyst offered by INS Pulawy (Łukasiewicz Research Network – New Chemical Syntheses Institute) [179].

Primary steam reforming catalysts	Secondary steam reforming and partial oxidation of natural gas catalysts
G-0117 G-0117K G-0217 G-0221 series	G-0110 G-0210 series
	

was performed by Ref. [185]. Wet impregnation was used to manufacture Ni-based (over MgO and Al_2O_3) catalysts, which were then impregnated in microreactors. Different steam-to-carbon, or s/c, ratios, temperatures, and air pressures were used to test the catalytic activity. While methane conversion was higher when the temperature was higher, it did not significantly increase when the steam-to-carbon ratio was higher. High methane and natural gas conversions were achieved with the Ni/Al_2O_3 catalyst in spite of its rapid deactivation. On the other hand, the Ni/[186]MgO catalyst for the SMR reaction did not appear to be deactivated and obtained the greatest hydrogen generation at 1073 K and s/c = 1.5. The pressure drop problem in reactors, particularly fixed-bed reactors, arises due to the resistance encountered by the fluid as it flows through the packed bed of catalyst particles. This resistance results from factors like particle size, shape, packing density, and bed porosity. As gas flow increases, discrepancies between experimental and numerical results also increase, highlighting the need for accurate modeling of particle properties and bed structure to ensure efficient reactor design and operation [186,187].

8. Methods of meeting the energy needs of the steam reforming process

Temperature is a key parameter of reforming as the SMR reaction is endothermic, heat should be delivered to the reactor during the process.

Among the methods of meeting the energy needs of the steam reforming process, the main ones can be distinguished: direct firing of fuel, waste heat recovery, integration with combined heat and power (CHP) systems, solar-assisted steam reforming, electric heating, partial oxidation (POX) integration, catalyst and reactor innovations, using energy from products of SMR and internal reforming in fuel cells.

The advantages and disadvantages of these methods are described below.

8.1. Direct firing of fuel

Meeting the energy needs of steam reforming is not a simple problem, as energy is consumed on the active surface of the reactor, while heat must be delivered from the "outside" of the reactor – usually simply by burning part of the natural gas. this method was the most popular in typical industrial installations for the production of hydrogen on both a small and large scale, before it was important to pay attention to energy efficiency and whether the hydrogen was green [188].

To investigate the heat required for a steam reforming industrial plant and the resulting net emissions of greenhouse gases, a life cycle analysis of hydrogen generation using natural gas steam reforming was conducted [189]. The hydrogen plant's capacity was set at 1.5 million Nm^3 per day. As in a typical steam reformer, the natural gas is transformed into synthesis gas, the synthesis gas is shifted in both high and low temperature shift reactors and is finally purified using a pressure swing adsorption unit. Analysis of the system's energy balance shows that the natural gas feedstock provides most of the energy utilized. According to higher heating value standards, the hydrogen plant's energy efficiency is 89.3 % (defined as the energy received from the system divided by the energy in the natural gas input into the system, assuming excess heat in the steam is consumed for industrial processes in the plant). Comparing the lower heating value (LHV) of energy in hydrogen to all energy in the fossil fuel consumed by the system, the efficiency reaches 66 %, which means that for every MJ of fossil fuel used by the system throughout the course of its lifetime, 0.66 MJ of hydrogen is generated (on an LHV basis). This illustrates how the system uses more energy than it generates since natural gas is a non-renewable resource. This value includes the energy consumed upstream to generate the power needed to run the hydrogen plant as well as to manufacture and distribute the natural gas.

The advantages of direct fuel combustion include stability, predictability, and ease of temperature control. The disadvantage of this solution is additional carbon dioxide emission.

8.2. Waste heat recovery

To improve the energy efficiency of the process of hydrogen production via SMR steam reforming reactors could be combined with industrial processes such involving forging furnaces [190] (see Fig. 16) or slags retrieved from blast furnaces [191–194].

The idea to apply Thermochemical Heat Regeneration (THR) – as simple heat recovery is not easy and a large heat excess occurs. THR is essentially the conversion of high-enthalpy waste-heat into the lower-enthalpy chemical energy of a new synthetic fuel (LHV). SMR is an endothermic process that can be used to regenerate thermochemical heat. The reaction between methane and steam produces synthesis gas, a novel synthetic fuel with an LHV higher than that of methane (a combination of carbon monoxide and hydrogen). This gas too can be used in furnaces for industrial processes.

This method can be applied in various industry systems to increase SMR efficiency (assuming no waste of methane for heat delivery to the reactor) or general energy efficiency of an industrial plant. In Ref. [195] the results of experiments of thermochemical waste-heat recuperation are presented (Figs. 17 and 18), showing the possibility of using this method for energy balance in various industry systems where temperatures of waste heat are between 500 and 1000 °C and the steam-to-carbon ratio is close to 2. The best results were achieved at 700 °C with 85 % being the maximum recovery level of heat from exhaust gases.

Energy recovery can also be implemented for smaller systems [196]. describes the use of exhaust gas heat from a piston engine for the methanol steam reforming process.

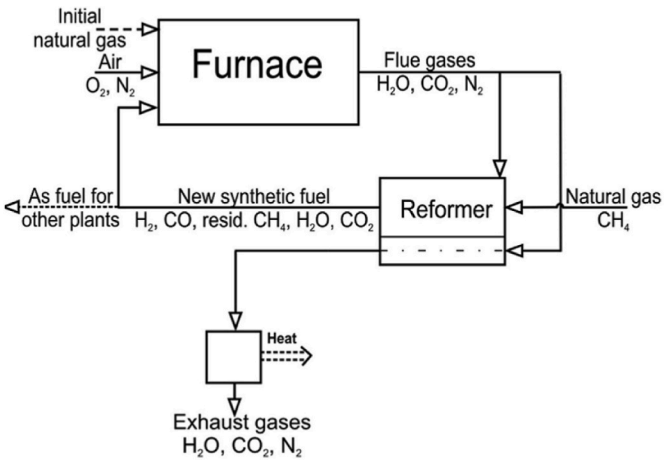


Fig. 16. Conceptual idea of the thermochemical heat regeneration for the industrial furnaces **(source:** [190]).

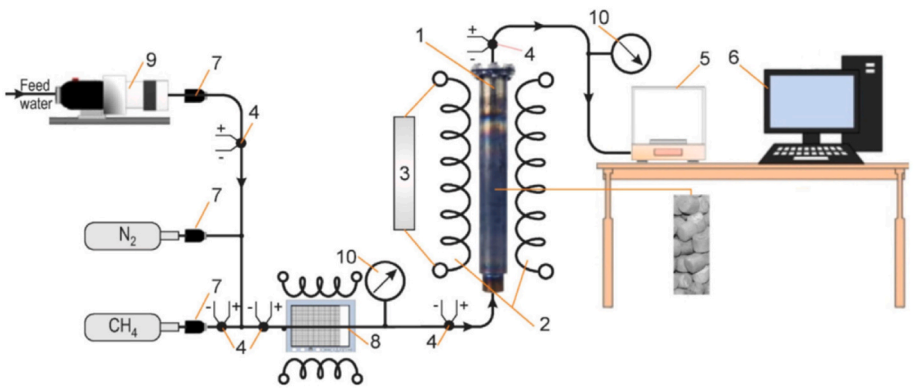


Fig. 17. Experimental setup of thermochemical waste-heat recuperation via SMR **(source:** [195]).

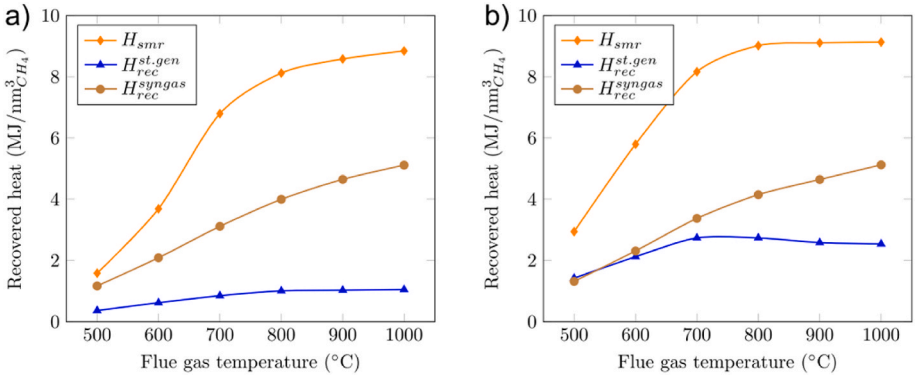


Fig. 18. Recovered heat in the reformer and steam generator **(source:** [195]).

The advantage of waste heat recovery is, as the name suggests, the use of heat that would otherwise be lost. This leads to increased efficiency of both processes and reduced carbon dioxide emissions. The disadvantage of this solution is that the size of the reforming installation is limited by the amount of heat from another installation. Another limitation is that the waste heat must be high temperature.

8.3. Integration with combined heat and power systems

Heat for the reforming process can also be obtained from CHP systems, which generate electricity and heat. Additionally, steam for reforming can also be obtained from such a system, which will further integrate these processes and increase the overall efficiency.

The disadvantage of this solution is that the size of the reforming installation is limited by the size of the CHP system.

The issues related to the integration of CHP systems with the reforming process have been described in Ref. [197]. The focus there was mainly on micro-CHP systems using PEMFC (Polymer Electrolyte Membrane Fuel Cells). Due to the fact that these fuel cells operate at low temperatures, it is impossible to carry out internal reforming (the last of the methods described in this chapter to cover the energy demand for the SMR process) using them.

In turn, a CHP system using a high-temperature PEM fuel cell (with a power of 50 kW), an ORC (Organic Rankine Cycle) system and steam methane reforming (Fig. 19) was presented in Ref. [198].

The maximum energy efficiency of this system was 60.98 % and electrical efficiency - 29.73 %.

8.4. Solar-assisted steam reforming

The heat for the endothermic reforming reaction can also be obtained from solar systems based on CSP (concentrated solar power) technology. In this way, heat for the reforming process and/or steam can be obtained.

The advantage of such a combination is the use of renewable energy in the reforming process and the reduction of fossil fuel (methane) consumption. The disadvantage of such a solution is that the process can only take place during the day.

In the case of using biogas in the solar-assisted steam reforming process, hydrogen can be obtained in a renewable way (see Fig. 20) [199].

Additionally, in the case of CO2 capture from the biogas reforming process, a negative carbon hydrogen production system (Fig. 20) can be obtained [199].

8.5. Electric heating

Electrical heating is another source of heat that can be used in the SMR process (see Fig. 21). This method of reforming is called electrical steam methane reforming (eSMR). The competitiveness of this method is compared with water electrolysis and conventional SMR in terms of CO_2 emissions in Ref. [200]. Since hydrogen produced continuously for industrial purposes was considered here, it came from the electricity grid and not from renewable energy sources. The carbon-CO_2 mineralization method was used in order to achieve zero emissions in eSMR.

The advantages of electric heating include the ease of controlling the temperature of the reforming process and the lack of additional CO_2 emission (in the case of electricity from renewable sources). The disadvantage of this solution may be the high cost of electricity.

8.6. Partial oxidation integration

Due to its exothermic course, partial oxidation of fuel (e.g. methane) can be combined with the endothermic reforming reaction. The

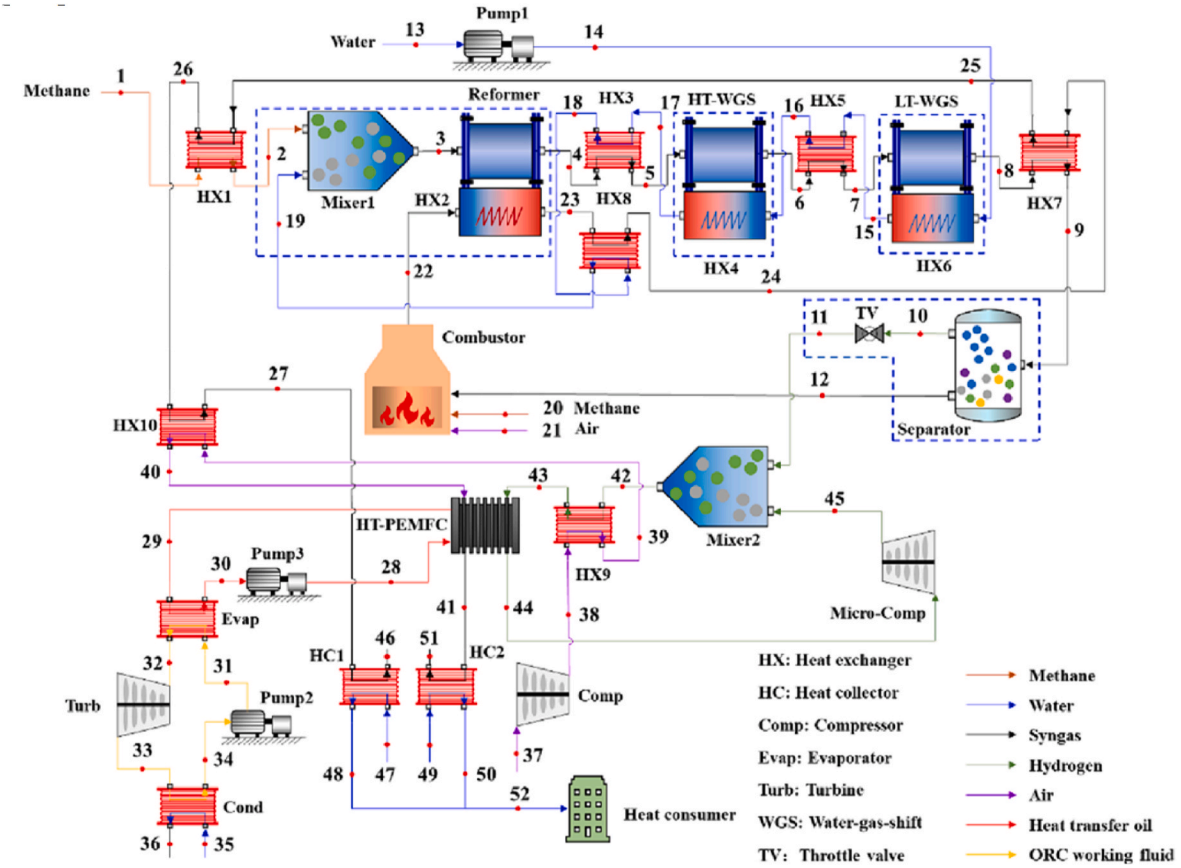


Fig. 19. Schematic diagram of a CHP system using a PEM fuel cell, ORC system and steam methane reforming **(source:** [198]**).**

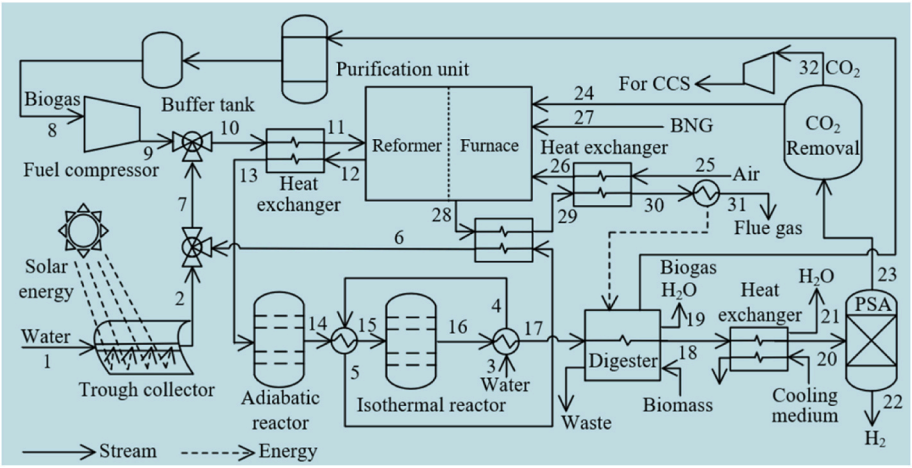


Fig. 20. Scheme of solar-assisted biogas steam reforming system with CO2 capture **(source:** [199]**).**

advantage of this solution is that it limits the demand for external energy, while the disadvantage is additional carbon dioxide emission. This solution was investigated by Ref. [201] where the integration of SMR and POX (Partial Oxidation of Methane) was achieved by combining the heat-compensating effects of DMR (Dry Methane Reforming) and POX with SMR, recycling large amounts of CO_2 into the DMR + POX section to maximize CO_2 reutilization. This integration leverages the exothermic nature of POX to offset the high heat demands of the endothermic SMR and DMR reactions, ensuring efficient syngas production with a desired H_2:CO ratio of 2:1. By recycling CO_2, the combined process significantly reduces raw material consumption and CO_2 emissions compared to conventional SMR. The advantages include a 31.2 % savings in energy consumption, 24.3 % utility cost savings, lower operating costs due to reduced raw material usage, and the production of syngas with lower CO_2 emissions. This makes the integrated process both economically and environmentally superior to the traditional SMR process.

8.7. *Catalyst and reactor innovations*

The use of suitably efficient catalysts can reduce the reaction activation energy and the temperature required for the reforming process to occur (and the lower temperature facilitates the use of waste heat –

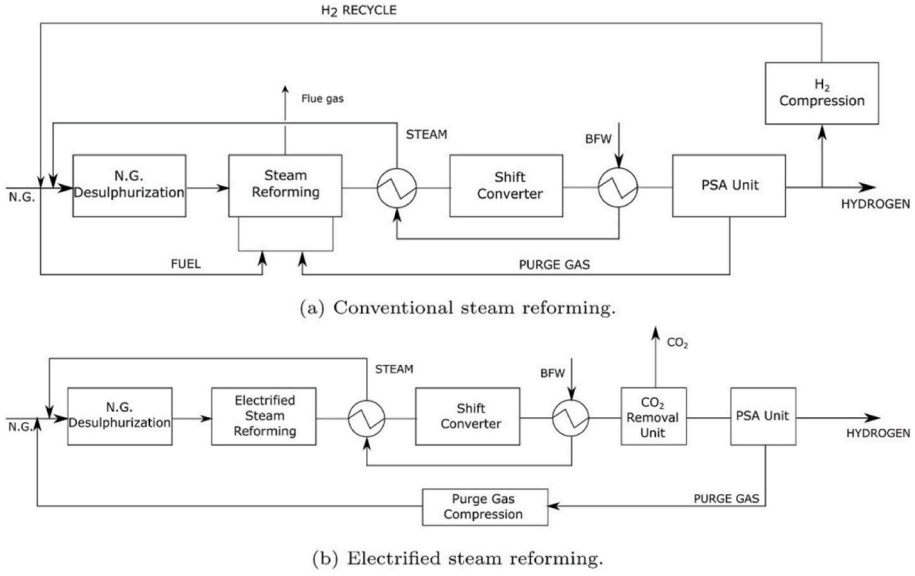


Fig. 21. Block diagrams of the conventional steam reforming and electrical steam methane reforming process **(source:** [200]**).**

second method). Another method may be to improve the design of reactors, e.g. by using microchannels or heat-exchanging reformers, etc. The advantage of such solutions is the increase in process efficiency and reduction of energy needs. The disadvantage of such solutions is the high investment costs.

The problem concerning heat distribution can be solved by the application of well heat transfer parameters of catalysts or cores of catalysts [202]. This however does not solve the general problem with the need to deliver heat for the process.

8.8. *Using energy from products of SMR*

Burning some of the reforming process products (hydrogen and carbon monoxide) to obtain heat for the process itself is possible, although less profitable than burning methane (first method – direct firing of fuel).

The disadvantages of this solution also include problems with flame stability in the case of too high hydrogen concentration in the fuel. This case was analyzed by Ref. [203] however solution required additional fuel input info off-gas from for combustion. Authors tried optimizing industrial plant to maximize hydrogen production and export steam flow rates and declared reaching satisfactory results, however results do not include measurable results in terms of cost or energy efficiency.

8.9. *Internal reforming in fuel cells*

Another solution is to combine the steam reforming process directly with internal steam reforming in the fuel cell [164]. As the electrochemical reaction in the fuel cell generates both electricity and heat, the requirement for heat in the endothermic reaction of steam reforming process can be easily covered [204], usually with excess, thus the fuel cell can still be considered a cogeneration unit even working with methane and steam as a fuel.

The disadvantage of this solution is that it can only be used for high-temperature fuel cells such as Solid Oxide Fuel Cell (SOFC) and Molten Carbonate Fuel Cell (MCFC).

The choice of the appropriate energy management for the steam methane reforming process depends on many aspects such as the size of the installation, the availability of energy sources (including waste or renewable energy), economic and environmental aspects, etc. The selection of the method of meeting the energy needs for the SMR process should be adapted to the specific conditions in which the installation will operate.

9. Methods of hydrogen purification after SMR

Syngas produced as a result of the steam reforming process, in addition to hydrogen, has a high carbon monoxide content, which is then converted to CO_2 in the WGS reactor in the presence of steam, releasing another portion of hydrogen. The resulting gaseous products contain a lot of hydrogen but also a significant amount of CO_2. Since adequate hydrogen purity is required for many industrial and sometimes energy applications, the process of purifying post-reforming syngas from CO_2 is carried out on a large scale.

According to Ref. [205], there are the following methods of capturing CO_2 from gases after SMR: pressure swing adsorption (PSA), amine methods (MEA - monoethanolamine, MDEA - methyldiethanolamine, aMDEA - activated methyldiethanolamine), the hot potassium carbonate method, physical solvents methods, membrane methods.

Pressure swing adsorption takes advantage of the fact that at high pressures certain gases can be captured by different adsorbents on their surface. After the pressure is decreased, the adsorbent releases the gas. Through repeated pressure changes, selected components of the gas can be captured and then released. The PSA method for purifying syngas after methane reforming is described in Refs. [206–208]. In turn, a comparison of different absorbents (HTC, CaO, Li4SiO4 and Li2ZrO3) under non-isothermal, non-isobaric and non-adiabatic operating conditions was performed in Ref. [209].

Amine CO_2 capture methods rely on the absorption of this gas by aqueous solutions of various amines through a chemical reaction resulting in the formation of soluble carbonate salts. Then, by heating the aqueous solution of these salts, this reaction is reversed and we obtain CO_2 and amines. In order to capture CO_2 from the syngas, it must be rinsed with an amine solution inside the absorber column. Amine methods for gas purification after SMR are described in Refs. [210–212].

Gases of various kinds can also be purified from CO_2 with an aqueous solution of potassium carbonate (25–35 % by weight) in a situation where the CO_2 will have a relatively high partial pressure [213]. This means that the concentration of this CO_2 in the gas to be cleaned must also be relatively high. This method is similar to the amine methods, with the difference being that the hot potassium carbonate method requires a much higher temperature [213]. An example of the application of this method to hydrogen purification after methane reforming is described in Ref. [214].

As mentioned before, CO_2 can be removed from the gases with the use of physical solvents. This method also uses the absorption phenomenon, but without chemical reactions [215].

The ideal conditions for such absorption are high CO_2 concentration in the gas (high partial pressure) and low temperature [215]. After absorbing significant amounts of CO_2, the solvent should undergo a regeneration process. Examples of commercial technologies using physical absorption are [215]: Selexol™, Rectisol™, Ipexol-2™, Fluor™, Purisol™, Sulfinol™ and Morphysorb™.

Removal of CO_2 from gas can also be performed using membrane technology. This technology hinges on the membrane only being permeable for selected components of the filtered medium. The purification of syngas after SMR in the case of membrane methods is based on the fact that the hydrogen permeates these membranes, which makes this method different from the others mentioned in this section. In other words, the syngas in this case is purified from hydrogen. In the case of SMR, there are solutions in which the reforming reactor itself is also a hydrogen separating membrane [216–218]. It is also possible to design an SMR membrane reactor whose membrane is permeable to CO_2 [219], thus separating it from the rest of gases.

Another possible method of removing CO_2 is carbon mineralization. This method converts CO2 into magnesium carbonates or calcium carbonates. This method is used in industrial processes and the resulting carbonates can be safely stored long-term. An example of the application of the method of coal mineralization from CO_2 formed in the eSMR process is presented in Ref. [200].

Some fuel cells, such as low-temperature proton-exchange membrane fuel cells (LT-PEMFC), are very sensitive to the presence of carbon monoxide (they tolerate CO content below 10 ppm) [220]. In order to apply SMR in such cases, it is necessary to combine several methods of hydrogen purification. For this purpose [220], uses a combination of a membrane reactor containing a Pd-composite membrane with a methanation catalyst. As a result, synthesis gas containing a high concentration of hydrogen but also small residues of other products of the reforming reaction (including CO) was subjected to the methanation reaction. Thus, the CO content of the gas leaving the system was 4 ppm with a hydrogen content of 97.7 %. An equally interesting way to achieve this goal may be to reduce CO_2 generation and inhibit the WGS reaction [221] and then use the methanation method to remove carbon monoxide.

10. Discussion

Table 5 below summarizes the results of the review described in this article. These results were summarized in terms of the conditions under which the reaction took place (pressure, temperature), type of catalyst used (or lack thereof), steam-to-carbon ratio (s/c), installation size, percentage of methane conversion and installation location.

In the available literature, the largest number of scientific articles is about the reforming process at the laboratory scale and with the use of various catalysts. The catalyst facilitates the course of the reforming process by reducing the activation energy of the reaction, thereby lowering the temperature at which this reaction takes place. The presented research highlights significant differences in the degree of methane conversion for different catalysts and different reaction temperatures. An important parameter in steam methane reforming is the steam to carbon ratio, S:C. Other parameters which affects the conversion and equilibrium composition of SMR process are temperature and pressure. However, increasing steam to carbon ratio only is not enough to achieve conversion of 99 %. The temperature must also be increased to achieve the desired methane conversion. Studies reported that the noble metal rare earth catalysts based were highly active, mechanical stable and less prone to coke. These catalysts allow operation at lower steam to the methane ratios. It enhanced process intensification methane conversion and also increased the hydrogen yield. In study [130], the catalyst exhibited high methane conversions even equal 100 % at 750 °C and high H_2 selectivity and excellent thermal stability. The total pressure affected on the conversion and stability i. e. with an increase in pressure the catalysts calcined at 650 °C and lower temperatures deactivated [132]. Therefore, the conversion and stability were the highest for the catalyst calcined at 850 °C. The research studies [134] indicated that $Ni_2Al_2O_5$ catalyst give optimal performance for methane steam reforming and limiting the formation of carbon in process. Fang et al. [136] confirmed that Ni/$Y_2Zr_2O_7$-GNC catalyst exhibited the highest activity and stability and also the most potent coke resistance among all others catalysts. In the above studies, it was found that active metallic Ni surface area determined about the activity and coke-resistance of the catalyst. In study [61], presented a comparison the reactivity of two complex Ni–perovskite powder catalyst systems:

Ni–$La_{0.6}Sr_{0.4}FeO_{3d}$ (LSF) and Ni–$SrTi_{0.7}Fe_{0.3d}$ (STF) to methane steam reforming process.

The STF catalyst exhibited high methane conversions. Iglesias et al. [138] presented that higher temperatures was suitable for methane conversion and CO selectivity. The increasing vapor/methane ratio caused an increase in hydrogen yield and also reduction carbon formation for catalysts at 600 °C. Nazari et al. [131] showed the effect of Cu and Zn metals content on different Ni/Al_2O_3 catalyst loads. The increment of Cu content enhanced the catalyst stability and increases the CO selectivity. The increment of Zn content caused CH_4 conversion, H_2 yield, and less selectivity to the CO production. Results showed that the 15 % load of Ni on Al_2O_3 support and addition of Cu and Zn loads of 1 % and 5 % enhanced the catalyst stability and activity. It also improved the catalyst performance in the hydrogen production as well. In work [99], presented the effect of Pt content for Ni/$MgAl_2O_4$ catalyst on the activity, selectivity and stability of the catalyst for at atmospheric and high pressure (10 bar) reactor pressure. It was noted that the effect of increasing the Pt doping in the Ni/$MgAl_2O_4$ caused the increase the degree of reduction of the catalyst. The research results showed that the Ni/$Y_2Ti_2O_7$ catalyst exhibited the highest catalytic activity and the best coke-resistant ability in the steam reforming of methane for hydrogen production among all of the catalysts [147]. Noh et al. [148] recommended a influence the addition of SiC to the Ni based catalysts with modified supports containing alumina. The addition of SiC enabled to improvement in the catalytic activity and coke resistance. Kim et al. [125] showed that the supported Ni catalysts about core/shell structures of Ni/Al_2O_3 and Ni/MgO–Al_2O_3 yielded a 97 % methane conversion with $H_2O/CH_4 = 2$ at 750 °C. However, the supported Ni catalysts about core/shell structures of CeO_2 were less activity and thermodynamic stability than the supported Ni catalysts about core/shell structures of Ni/Al_2O_3 and Ni/MgO–Al_2O_3. Izquierdo et al. [185] showed a high methane and natural gas conversions for the Ni/Al_2O_3 catalyst. Higher methane conversion was obtained at higher temperatures.

11. Conclusions

Most of the studies available in the literature on non-catalytic SMR concern mainly theoretical issues. The absence of a catalyst results in the reforming taking place at much higher temperatures than is the case with catalytic reforming and the chemical reactions involved following a different course.

SMR is catalyzed by various non-noble and noble metals. The Ni based catalysts are attracted by their high activity for tar reduction, low cost and easy regeneration.

The deactivation caused by coke deposition and metal sintering is a serious problem. Therefore, modified Ni-based catalysts are now most frequently used for steam methane catalytic reforming.

The article describes methods of meeting the energy needs of the steam reforming process (direct firing of fuel, waste heat recovery, integration with CHP systems, solar-assisted steam reforming, electric heating, partial oxidation integration, catalyst and reactor innovations, using energy from products of SMR and internal reforming in fuel cells) and methods of purifying the hydrogen obtained (pressure swing

Table 5
Summary of research installations for the SMR process.

Source	Working temperature and pressure T [°C]	Catalyst	s/c	Installation size	Methane conversion	Location
[130]	T = 650–900 P = 1 bar	$Ni/SiO_2Al_2O_3$ (5–8±1.2 nm) Ni (5 wt%) SiO_2 (10 wt%) $Ni/SiO_2Al_2O_3$ (18–20 ± 3.1 nm) Ni (5 wt%) SiO_2 (10 wt%)	$CH_4:H_2O$ = 1:1	Laboratory scale	100 % CH_4 (T = 750, P = 1 bar) 75 % CH_4 (T = 750, P = 1 bar) 100 % CH_4 (T = 900, P = 1 bar)	Qatar
[132]	T = 600 P = 1–10 bar	$Ni/MgAl_2O_4$ (0.25–0.30 mm) Ni (15 wt%) Mg (30 wt%) Al_2O_4 (70 wt%)	$CH_4:H_2O:N_2$ = 1:5:1	Laboratory scale	40–50 % CH_4 (T = 600, P = 1 bar) 35–45 % CH_4 (T = 600, P = 5 bar) 35–45 % CH_4 (T = 600, P = 10 bar)	India
[134]	T = 700 P = 1 bar	$NiAl_4O_7$ (8–10 nm) Ni (20 wt%) $NiAl_2O_4$ Ni (34 wt%) (6–7 nm) $Ni_2Al_2O_5$ (3–4 nm) Ni (45 wt%)	$H_2O:CH_4:N2$ = 2.4:1:3.4	Laboratory scale	$NiAl_4O_7$ not active; $NiAl_2O_4$ 78 % CH_4 (T = 700, P = 1 bar); $Ni_2Al_2O_5$ 82 % CH_4 (T = 700, P = 1 bar)	USA
[136]	T = 550-800 P = 1, 20 atm	$Ni/Y_2Zr_2O_7$_x x = HT,CP,GNC Ni (9.8 wt%; x = HT) Ni (9.7 wt%; x = CP) Ni (9.8 wt%; x = GNC)	$H_2O:CH_4$ = 2:1	Laboratory scale	$Ni/Y_2Zr_2O_7$_HT 83 % CH_4 (T = 800, P = 1atm); $Ni/Y_2Zr_2O_7$_CP 60 % CH_4 (T = 800, P = 1atm); $Ni/Y_2Zr_2O_7$_GNC 98 % CH_4 (T = 800, P = 1atm)	China
[61]	T = 550-600 P = 1 bar	$Ni\text{-}La_{0.6}Sr_{0.4}FeO_{3-\delta}$ (50 nm) Ni (10 wt%) $Ni\text{-}SrTi_{0.7}Fe_{0.3}O_{3-\delta}$ (25 nm) Ni (10 wt%)	$H_2O:CH_4$ = 1:1	Laboratory scale	$Ni\text{-}SrTi_{0.7}Fe_{0.3}O_{3-\delta}$ 90 % CH_4 (T = 600, P = 1atm)	Austria
[138]	T = 500-900	$Ni/Ce_{0.95}M_{0.05}O_{2-d}$ (M = Zr,Pr,La) (17–35 nm) Ni (5 wt%) Zr,Pr,La (5 at.%)	$H_2O:CH_4$ = 0.5:1; 1:1; 1.5:1; 3:1	Laboratory scale	5 % $Ni/CeZr_5$ 72 % CH_4 (T = 600)	Argentina

[131]	T = 700	$xNi\text{-}yCu\text{-}tZn/\gamma\text{-}Al_2O_3$
		x = 5,10,15;
		y = 1,3,5;
		t = 1,3,5
		Ni (15 wt%)
			$H_2:CH_4$ = 4:1	Laboratory scale	15Ni-1Cu-5Zn/γ-Al_2O_3 95 % CH_4 (T = 700)	Iran

[135]	T = 750-850	$Ni/MgAl_2O_4$ (14.9 nm) $Ni/Mg/\gamma\text{-}Al_2O_3$ (21.9 nm) $NiMgAlO_x$ (42.1 nm)	$H_2O:CH_4$ = 1.24:1	Laboratory scale	74 % CH_4 (T = 850)	Spain
[99]	T = 600 P = 1-10	xPt/ $Ni/MgAl_2O_4$ Ni (15 wt%) Pt (0.01–1.0 wt%)	$CH_4:H_2O:N_2$ = 1:5:1	Laboratory scale	19–63 % CH_4 (T = 600, P = 1 atm)	India
[147]	T = 600-800	$Ni/Ln_2Ti_2O_7$ (Ln = La, Pr, Sm, Y) (16.6–17.5 nm) Ni (7.1 wt%) $La_2Ti_2O_7$ Ni (6.9 wt%) $Pr_2Ti_2O_7$ Ni (7.1 wt%) $Sn_2Ti_2O_2$ Ni (7.0 wt%) $Y_2Ti_2O_7$	H_2O: CH_4 = 2:1	Laboratory scale	$Ni/Y_2Ti_2O_7$ 85 % CH_4 (T = 800)	China
[143]	T = 800 P = 15–30 bar	$Ni/La_2Zr_2O_7$ Ni (10.5 wt%) $Ni/Pr_2Zr_2O_7$ Ni (18.5 wt%) $Ni/Sm_2Zr_2O_7$ Ni (7.8 wt%) $Ni/Y_2Zr_2O_7$ Ni (5.2 wt%)	$H_2O:CH_4$ = 2:1	Laboratory scale	<99 % CH_4	China

(*continued on next page*)

Table 5 (*continued*)

Source	Working temperature and pressure T [°C]	Catalyst	s/c	Installation size	Methane conversion	Location
[146]	T = 500-800	$NiO/\gamma\text{-}Al_2O_3$ (11.4–14.2 nm) Ni (5 wt%)	$H_2O:CH_4$ = 3:1	Laboratory scale	5Ni/γ-Al_2O_3 90 % CH_4 (T = 700)	Algeria
[148]	T = 650-850 P = 1 bar	$NiAl_2O_4$ Ni (10 wt%)	$CH_4:H_2O:H_2$ = 1:3:0.5	Laboratory scale	97.9 % CH_4 (T = 650); 81.3 % CH_4 (T = 850)	South Korea
[137]	T = 600-800	$Ni/CaZrO_3/\alpha\text{-}Al_2O_3$ Ni (15 wt%) $CaZrO_3$ (15 wt%)	$H_2O:CH_4:Ar$ = 1:1:3; 1:3:6; 3:1:6	Laboratory scale	15Ni/15$CaZrO_3$/α-Al_2O_3 67 % CH_4 (T = 700)	Thailand
[48]	T = 500 P = 30 bar	$Ni\text{-}X/Al_2O_3$ X = Ru,Rh,Ir (7–30 nm) Ni (2.89 wt%) Ni/Al_2O_3 Ni (2.91 wt%) $Ni\text{-}Ir/Al_2O_3$ Ni (2.92 wt%) $Ni\text{-}Rh/Al_2O_3$ Ni (2.92 wt%) $Ni\text{-}Ru/Al_2O_3$	$CH_4:H_2O:H_2$ = 2:8:0.8	Laboratory scale	no data	Denmark
[133]	T = 500-800	$xNi\text{-}yLa/\alpha\text{-}Al_2O_3$ (x/y = 7/3; 8/2; 9/1) xNi-yLa (10 wt%)	$CH_4:H_2O$ = 1:3	Laboratory scale	7Ni-3La/α-Al_2O_3 87 % CH_4 (T = 600)	Algeria
[145]	T = 730-830 P = 1 atm	mNi/xLa-Si (m = 7.5; 12.5; 15; 17.5; 22.5) (x = 0; 1.0; 2.0; 3.0; 5.0; 7.0) (14–20 nm) Ni (17.5 wt%) Ni/0La-Si Ni/1.0La-Si Ni/2.0La-Si Ni/3.0La-Si Ni/5.0La-Si Ni/7.0La-Si	$CH_4:CO_2:H_2O$ = 1:0.4:0.8	Laboratory scale	17.5Ni/3.0La-Si 85 % CH_4 (T = 800)	China
[149]	T = 600-700	Ni/Al_2O_3 (6–12 nm) Ni (10 wt%)	$CH_4:CO_2$ = 1:0.48 H_2O/CH_4 = 1.2:0.48 3.5:0.48 6.1:0.48	Laboratory scale	10 wt%Ni/ Al_2O_3 99 % CH_4 (T = 700)	Romania
[166]	T = 500–750 P = 1 bar	No data	2, 2.5, 3, 3.5	Laboratory scale	99 % (T = 750)	Poland
[222]	T = 700 P = 1 atm	$Ni/S/Al_2O_3$ Ni (7; 14; 28 wt%)	$H_2O:CH_4$:air = 2.5:1:3.5	Laboratory scale	99 % CH_4	China/Japan
[223]	T = 600 P = 0.987 atm	$Ni/Mg/Al/Al_2O_3$ Ni (7 wt%) $Ni/Zn/Al/Al_2O_3$ Ni (8.6 % wt.)	$H_2O:CH_4$ = 2:1	Laboratory scale	>80 % CH_4 >75 % CH_4	South Korea
[224]	T = 650 P = 1 atm	Ni/SiO_2 Ni (15 wt%)	$H_2O:CH_4$ = 2:1	Laboratory scale	89 % CH_4	Brazil
[225]	T = 635 P = 1 atm T = 660 P = 1 atm T = 699 P = 1 atm T = 699 P = 1 atm T = 800 P = 1 atm T = 745 P = 1 atm	Rh/CeO_2 Rh (1.5 wt%) Rh/Al_2O_3 Rh (1.5 wt%) Ru/CeO_2 Ru (1.5 wt%) Ru/Al_2O_3 Ru (1.5 wt%) Pt/CeO_2 Pt (1.1 wt%) Pt/Al_2O_3 Pt (1.1 wt%)	$H_2O:CH_4$ = 3.5:1	Laboratory scale	>99 % CH_4 >99 % CH_4 98.8 % CH_4 >99 % CH_4 <80 % CH_4 >99 % CH_4	Poland
[226]	T = 550-750 P = 1 atm	HT-xNi x = 2.79–58.66 wt% Ni	$CH_4:CO_2:Ar$ = 1:1:8	Laboratory scale	>80 % CH_4	Poland
[125]	T = 750 P = 1 atm	Ni/Al_2O_3 $Ni/MgAl_2O_4$ Ni/CeO_2 $Ni/Ce_{0.4}Zr_{0.6}O_2$ Ni (10 wt%)	$H_2O:CH_4$ = 2:1	Laboratory scale	97 % CH_4 97.2 % CH_4 89 % CH_4 91 % CH_4	Republic of Korea
[185]	T = 700-800 P = 1 atm	Ni/Al_2O_3 Ni (18 wt%) Ni/MgO Ni (17.02 wt%) Pt/Al_2O_3 Pt (0.79 wt%)	$CH_4:CO_2:Ar$ = 1:1.5:2.0	Laboratory scale	99 % CH_4	Spain

(*continued on next page*)

Table 5 (*continued*)

Source	Working temperature and pressure T [°C]	Catalyst	s/c	Installation size	Methane conversion	Location
[227]	T = 800-950 P = 1 atm	Pd/Al_2O_3 Pd (0.31 wt%) (NiAl)600 Ni (1.3 wt%) (0.01 Pt(NiAl) 600)300 Ni (0.9 wt%) (0.3 Pt(NiAl)600)300 Ni (0.2 wt%) (NiAl)800 Ni (1.7 wt%) (0.01 Pt(NiAl) 800)300 Ni (1.3 wt%) (0.3 Pt(NiAl)800)300 Ni (0.6 wt%)	7:1	Laboratory scale	No data	Brazil
[228]	T = 750 P = 1 atm	$Ni/MgAl_2O_4$ Ni (14 wt%)	$H_2O:CH_4$ = 2.5:1	Laboratory scale	<99 % CH_4	Denmark

adsorption, amine methods, the hot potassium carbonate method, physical solvents methods and membrane methods). The use of membrane reforming reactors, which can simultaneously purify the hydrogen produced through reforming, seem to be prospective technologies for the development of SMR. eSMR is also of great interest, particularly when the electricity used for this comes from renewable sources. To cover the energy needs of the reforming process, it is also beneficial to use waste heat from another process.

In order to use SMR to power low-temperature fuel cells, an interesting solution is to use the methanation method of post-reforming gases (without previously captured CO_2) in order to obtain sufficiently low carbon monoxide content. An equally interesting way to achieve this goal may be to reduce CO_2 generation and inhibit the WGS reaction [221] and then use the methanation method to remove carbon monoxide.

The article also presents a mathematical description of the reactions occurring during steam reforming of methane.

CRediT authorship contribution statement

Lukasz Szablowski: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Malgorzata Wojcik:** Writing – original draft, Resources, Methodology, Investigation, Formal analysis. **Olaf Dybinski:** Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2025.134540.

Data availability

No data was used for the research described in the article.

References

[1] McCay MH, Shafiee S. 22 - hydrogen: an energy carrier. Future energy. In: Improved, sustainable and clean options for our planet. third ed. Elsevier; 2020. p. 475–93. https://doi.org/10.1016/B978-0-08-102886-5.00022-0.
[2] Zgonnik V. The occurrence and geoscience of natural hydrogen: a comprehensive review. Earth Sci Rev 2020;203. https://doi.org/10.1016/j.earscirev.2020.103140.
[3] IEA. Global hydrogen review 2022. International Energy Agency; 2022.
[4] IRENA. Geopolitics of the energy transformation: the hydrogen factor. Abu Dhabi: International Renewable Energy Agency; 2022.
[5] Hydrogen Council in collaboration with McKinsey & Company. Hydrogen insights 2022. Hydrogen Council; 2022.
[6] Bird & Bird. International green hydrogen report 2022. Bird & Bird LLP; 2022.
[7] Kochański M, Korczak K, Dybiński O, Kwas M, Osipowicz K, Patejuk A, et al. The technical and economical perspectives for the production and storage of hydrogen in Poland. Acta Innovations 2013;8:50–62.
[8] Wongsakulphasatch S, Noppakun N, Chimpae S, Nimmas T, Kim-Lohsoontorn P, Powell J, et al. Intensified processes of steam reforming and their materials for hydrogen production. New Dimensions in Production and Utilization of Hydrogen 2020. https://doi.org/10.1016/B978-0-12-819553-6.00005-2.
[9] Muradov NZ, Veziroğlu TN. From hydrocarbon to hydrogen-carbon to hydrogen economy. Int J Hydrogen Energy 2005;30. https://doi.org/10.1016/j.ijhydene.2004.03.033.
[10] Cherif A, Nebbali R, Sheffield JW, Doner N, Sen F. Numerical investigation of hydrogen production via autothermal reforming of steam and methane over Ni/Al2O3 and Pt/Al2O3 patterned catalytic layers. Int J Hydrogen Energy 2021;46. https://doi.org/10.1016/j.ijhydene.2021.04.032.
[11] Settar A, Mansouri Z, Nebbali R, Madani B, Abboudi S. Impact of Ni-based catalyst patterning on hydrogen production from MSR: external steam reformer modelling. Int J Hydrogen Energy 2019. https://doi.org/10.1016/j.ijhydene.2018.09.171.
[12] Cherif A, Nebbali R, Lee CJ. Design and multiobjective optimization of membrane steam methane reformer: a computational fluid dynamic analysis. Int J Energy Res 2022;46. https://doi.org/10.1002/er.7750.
[13] Mundhwa M, Thurgood CP. Numerical study of methane steam reforming and methane combustion over the segmented and continuously coated layers of catalysts in a plate reactor. Fuel Process Technol 2017;158. https://doi.org/10.1016/j.fuproc.2016.12.002.
[14] Fu WB, Hou LY, Zhong BJ, Ma FH, Qin C, Li C Le. An analysis of the ignition of premixed gases by a hot spherical surface with catalytic reforming reaction. Fuel 2003;82. https://doi.org/10.1016/S0016-2361(02)00293-4.
[15] Souentie S, Athanasiou M, Niakolas DK, Katsaounis A, Neophytides SG, Vayenas CG. Mathematical modeling of Ni/GDC and Au-Ni/GDC SOFC anodes performance under internal methane steam reforming conditions. J Catal 2013;306. https://doi.org/10.1016/j.jcat.2013.06.015.
[16] Sciazko A, Komatsu Y, Brus G, Kimijima S, Szmyd JS. A novel approach to the experimental study on methane/steam reforming kinetics using the Orthogonal Least Squares method. J Power Sources 2014;262. https://doi.org/10.1016/j.jpowsour.2014.03.097.
[17] Avetisov AK, Rostrup-Nielsen JR, Kuchaev VL, Bak Hansen JH, Zyskin AG, Shapatina EN. Steady-state kinetics and mechanism of methane reforming with

steam and carbon dioxide over Ni catalyst. J Mol Catal Chem 2010;315. https://doi.org/10.1016/j.molcata.2009.06.013.
[18] Froment GF, Xu J. Methane steam reforming , methanation and water-gas shift : 1 . Intrinsic kinetics. AIChE J 1989;35.
[19] Wu Z, Guo Z, Yang J, Wang Q. Numerical investigation of methane steam reforming in packed bed reactor with internal helical heat fins. Energy 2023;278. https://doi.org/10.1016/j.energy.2023.127988.
[20] Wójcik M, Szabłowski Ł, Dybiński O. Comparison of mathematical models of steam methane reforming process for the needs of fuel cells. Int J Hydrogen Energy 2024;52. https://doi.org/10.1016/j.ijhydene.2023.08.293.
[21] Rostrup-Nielsen JR, Hansen JB. Chapter 4 - steam reforming for fuel cells. In: Shekhawat D, Spivey JJ, Berry DA, editors. Fuel cells: technologies for fuel processing. Amsterdam: Elsevier; 2011. p. 49–71. https://doi.org/10.1016/B978-0-444-53563-4.10004-5.
[22] Karim GA, Metwally MM. A kinetic investigation of the reforming of natural gas for the production of hydrogen. Int J Hydrogen Energy 1980;5. https://doi.org/10.1016/0360-3199(80)90073-7.
[23] Savchenko VI, Zimin YS, Nikitin AV, Sedov IV, Arutyunov VS. Non-catalytic steam reforming of C1–C4 hydrocarbons. Petrol Chem 2021;61. https://doi.org/10.1134/S0965544121070021.
[24] Xiong Z, Fang Z, Jiang L, Han H, He L, Xu K, et al. Comparative study of catalytic and non-catalytic steam reforming of bio-oil: importance of pyrolysis temperature and its parent biomass particle size during bio-oil production process. Fuel 2022;314. https://doi.org/10.1016/j.fuel.2021.122746.
[25] Latifi M, Berruti F, Briens C. Non-catalytic and catalytic steam reforming of a bio-oil model compound in a novel "jiggle Bed" Reactor. Fuel 2014;129. https://doi.org/10.1016/j.fuel.2014.03.053.
[26] Vourliotakis G, Skevis G, Founti MA. Detailed kinetic modelling of non-catalytic ethanol reforming for SOFC applications. Int J Hydrogen Energy 2009;34. https://doi.org/10.1016/j.ijhydene.2009.07.007.
[27] Pedersen-Mjaanes H, Chan L, Mastorakos E. Hydrogen production from rich combustion in porous media. Int J Hydrogen Energy 2005;30. https://doi.org/10.1016/j.ijhydene.2004.05.006.
[28] Richter A, Seifert P, Compart F, Tischer P, Meyer B. A large-scale benchmark for the CFD modeling of non-catalytic reforming of natural gas based on the Freiberg test plant HP POX. Fuel 2015;152. https://doi.org/10.1016/j.fuel.2014.12.004.
[29] Guiberti TF, Garnier C, Scouflaire P, Caudal J, Labegorre B, Schuller T, et al. Experimental and numerical analysis of non-catalytic partial oxidation and steam reforming of CH4/O2/N2/H2O mixtures including the impact of radiative heat losses. Int J Hydrogen Energy 2016;41. https://doi.org/10.1016/j.ijhydene.2016.03.009.
[30] Espinoza L, Guerrero F, Ripoll N, Toledo M, Guerrero L, Carvajal A, et al. Syngas production by non-catalytic reforming of biogas with steam addition under filtration combustion mode. Int J Hydrogen Energy 2018;43. https://doi.org/10.1016/j.ijhydene.2018.06.136.
[31] Dybiński O, Milewski J, Szczęśniak A, Martsinchyk A, Szabłowski Ł. Experimental investigation of porous anode degradation of a molten carbonate fuel cell fed with direct fermentation product composed of bioethanol. Int J Hydrogen Energy 2023. https://doi.org/10.1016/J.IJHYDENE.2023.04.065.
[32] Dybiński O, Milewski J, Szabłowski Ł, Szczęśniak A, Martinchyk A. Methanol, ethanol, propanol, butanol and glycerol as hydrogen carriers for direct utilization in molten carbonate fuel cells. Int J Hydrogen Energy 2023. https://doi.org/10.1016/J.IJHYDENE.2023.05.091.
[33] Zhou X, Chen C, Wang F. Multi-dimensional modeling of non-catalytic partial oxidation of natural gas in a high pressure reformer. Int J Hydrogen Energy 2010;35. https://doi.org/10.1016/j.ijhydene.2009.12.063.
[34] Förster T, Voloshchuk Y, Richter A, Meyer B. 3D numerical study of the performance of different burner concepts for the high-pressure non-catalytic natural gas reforming based on the Freiberg semi-industrial test facility HP POX. Fuel 2017;203. https://doi.org/10.1016/j.fuel.2017.03.089.
[35] Blom PWE, Basson GW. Non-catalytic plasma-arc reforming of natural gas with carbon dioxide as the oxidizing agent for the production of synthesis gas or hydrogen. Int J Hydrogen Energy 2013;38. https://doi.org/10.1016/j.ijhydene.2013.03.042.
[36] Xu Y, Dai Z, Li C, Li X, Zhou Z, Yu G, et al. Numerical simulation of natural gas non-catalytic partial oxidation reformer. Int J Hydrogen Energy 2014;39. https://doi.org/10.1016/j.ijhydene.2014.03.204.
[37] Voloshchuk Y, Vascellari M, Hasse C, Meyer B, Richter A. Numerical study of natural gas reforming by non-catalytic partial oxidation based on the Virtuhcon Benchmark. Chem Eng J 2017;327. https://doi.org/10.1016/j.cej.2017.06.061.
[38] Guo W, Wu Y, Dong L, Chen C, Wang F. Simulation of non-catalytic partial oxidation and scale-up of natural gas reformer. Fuel Process Technol 2012;98. https://doi.org/10.1016/j.fuproc.2012.01.019.
[39] Savchenko VI, Zimin YS, Nikitin AV, Sedov IV, Arutyunov VS. Utilization of CO2in non-catalytic dry reforming of C1-C4hydrocarbons. J CO2 Util 2021;47. https://doi.org/10.1016/j.jcou.2021.101490.
[40] Pashchenko D, Makarov I. Carbon deposition in steam methane reforming over a Ni-based catalyst: experimental and thermodynamic analysis. Energy 2021;222. https://doi.org/10.1016/j.energy.2021.119993.
[41] Jang WJ, Shim JO, Kim HM, Yoo SY, Roh HS. A review on dry reforming of methane in aspect of catalytic properties. Catal Today 2019. https://doi.org/10.1016/j.cattod.2018.07.032.
[42] Nieva MA, Villaverde MM, Monzón A, Garetto TF, Marchi AJ. Steam-methane reforming at low temperature on nickel-based catalysts. Chem Eng J 2014;235. https://doi.org/10.1016/j.cej.2013.09.030.
[43] Kikuchi E, Tanaka S, Yamazaki Y, Morita Y. Steam reforming of hydrocarbons on noble metal catalysts (Part 1). Bull Jpn Petrol Inst 1974;16. https://doi.org/10.1627/jpi1959.16.95.
[44] Rostrup-Nielsen JR, Bak Hansen JH. CO2-reforming of methane over transition metals. J Catal 1993;144. https://doi.org/10.1006/jcat.1993.1312.
[45] Qin D, Lapszewicz J. Study of mixed steam and CO2 reforming of CH4 to syngas on MgO-supported metals. Catal Today 1994;21. https://doi.org/10.1016/0920-5861(94)80179-7.
[46] Jones G, Jakobsen JG, Shim SS, Kleis J, Andersson MP, Rossmeisl J, et al. First principles calculations and experimental insight into methane steam reforming over transition metal catalysts. J Catal 2008;259. https://doi.org/10.1016/j.jcat.2008.08.003.
[47] Wei J, Iglesia E. Isotopic and kinetic assessment of the mechanism of reactions of CH 4 with CO2 or H2O to form synthesis gas and carbon on nickel catalysts. J Catal 2004;224:370–83. https://doi.org/10.1016/j.jcat.2004.02.032.
[48] Morales-Cano F, Lundegaard LF, Tiruvalam RR, Falsig H, Skjøth-Rasmussen MS. Improving the sintering resistance of Ni/Al2O3 steam-reforming catalysts by promotion with noble metals. Appl Catal Gen 2015;498. https://doi.org/10.1016/j.apcata.2015.03.016.
[49] Nawfal M, Gennequin C, Labaki M, Nsouli B, Aboukaïs A, Abi-Aad E. Hydrogen production by methane steam reforming over Ru supported on Ni-Mg-Al mixed oxides prepared via hydrotalcite route. Int J Hydrogen Energy 2015;40. https://doi.org/10.1016/j.ijhydene.2014.09.166.
[50] Yoon Y, Kim H, Lee J. Enhanced catalytic behavior of Ni alloys in steam methane reforming. J Power Sources 2017;359. https://doi.org/10.1016/j.jpowsour.2017.05.076.
[51] Álvarez M A, Bobadilla LF, Garcilaso V, Centeno MA, Odriozola JA. CO2 reforming of methane over Ni-Ru supported catalysts: on the nature of active sites by operando DRIFTS study. J CO2 Util 2018;24. https://doi.org/10.1016/j.jcou.2018.01.027.
[52] Luisetto I, Sarno C, De Felicis D, Basoli F, Battocchio C, Tuti S, et al. Packed and monolithic reactors for the dry reforming of methane: Ni supported on γ-Al2O3 promoted by Ru. Adv Sci Lett 2017;23. https://doi.org/10.1166/asl.2017.9085.
[53] Wysocka I, Hupka J, Rogala A. Catalytic activity of nickel and ruthenium–nickel catalysts supported on SiO2, ZrO2, Al2 O3, and MgAl2 O4 in a dry reforming process. Catalysts 2019;9. https://doi.org/10.3390/catal9060540.
[54] Zhou H, Zhang T, Sui Z, Zhu YA, Han C, Zhu K, et al. A single source method to generate Ru-Ni-MgO catalysts for methane dry reforming and the kinetic effect of Ru on carbon deposition and gasification. Appl Catal B Environ 2018;233. https://doi.org/10.1016/j.apcatb.2018.03.103.
[55] Tsodikov MV, Kurdymov SS, Konstantinov GI, Murzin VY, Bukhtenko OV, Maksimov YV. Core-shell bifunctional catalyst for steam methane reforming resistant to H2S: activity and structure evolution. Int J Hydrogen Energy 2015;40. https://doi.org/10.1016/j.ijhydene.2015.01.016.
[56] Konstantinov GI, Kurdyumov SS, Maksimov YV, Bukhtenko OV, Tsodikov MV. Hydrogen sulfide-resistant bifunctional catalysts for the steam reforming of methane: activity and structural evolution. Catalysis in Industry 2018;10. https://doi.org/10.1134/S2070050418010063.
[57] Theofanidis SA, Galvita VV, Sabbe M, Poelman H, Detavernier C, Marin GB. Controlling the stability of a Fe–Ni reforming catalyst: structural organization of the active components. Appl Catal B Environ 2017;209. https://doi.org/10.1016/j.apcatb.2017.03.025.
[58] Xu L li, Wen H, Jin X, Bing Q ming, Liu J yao. DFT study on dry reforming of methane over Ni 2 Fe overlayer of Ni(1 1 1) surface. Appl Surf Sci 2018;443. https://doi.org/10.1016/j.apsusc.2018.02.268.
[59] Ray K, Sandupatla AS, Deo G. Activity and stability descriptors of Ni based alloy catalysts for dry reforming of methane: a density functional theory study. Int J Quant Chem 2021;121. https://doi.org/10.1002/qua.26580.
[60] Djaidja A, Messaoudi H, Kaddeche D, Barama A. Study of Ni-M/MgO and Ni-M-Mg/Al (M=Fe or Cu) catalysts in the CH4-CO2 and CH4-H2O reforming. Int J Hydrogen Energy 2015;40. https://doi.org/10.1016/j.ijhydene.2014.12.106.
[61] Thalinger R, Gocyla M, Heggen M, Dunin-Borkowski R, Grünbacher M, Stöger-Pollach M, et al. Ni-perovskite interaction and its structural and catalytic consequences in methane steam reforming and methanation reactions. J Catal 2016;337. https://doi.org/10.1016/j.jcat.2016.01.020.
[62] Tomishige K, Li D, Tamura M, Nakagawa Y. Nickel-iron alloy catalysts for reforming of hydrocarbons: preparation, structure, and catalytic properties. Catal Sci Technol 2017;7. https://doi.org/10.1039/c7cy01300k.
[63] Song Z, Wang Q, Guo C, Li S, Yan W, Jiao W, et al. Improved effect of Fe on the stable NiFe/Al2O3 catalyst in low-temperature dry reforming of methane. Ind Eng Chem Res 2020;59. https://doi.org/10.1021/acs.iecr.0c01204.
[64] Margossian T, Larmier K, Kim SM, Krumeich F, Müller C, Copéret C. Supported bimetallic NiFe nanoparticles through colloid synthesis for improved dry reforming performance. ACS Catal 2017;7. https://doi.org/10.1021/acscatal.7b02091.
[65] Kim SM, Abdala PM, Margossian T, Hosseini D, Foppa L, Armutlulu A, et al. Cooperativity and dynamics increase the performance of NiFe dry reforming catalysts. J Am Chem Soc 2017;139. https://doi.org/10.1021/jacs.6b11487.
[66] Theofanidis SA, Galvita VV, Poelman H, Marin GB. Enhanced carbon-resistant dry reforming Fe-Ni catalyst: role of Fe. ACS Catal 2015;5. https://doi.org/10.1021/acscatal.5b00357.
[67] Gunduz-Meric G, Kaytakoglu S, Degirmenci L. Ni,Co/SiO2 and Ni/SiO2,Co bimetallic microsphere catalysts indicating high activity and stability in the dry reforming of methane. React Kinet Mech Catal 2020;129. https://doi.org/10.1007/s11144-019-01708-4.

[68] Katheria S, Deo G, Kunzru D. Rh-Ni/MgAl2O4 catalyst for steam reforming of methane: effect of Rh doping, calcination temperature and its application on metal monoliths. Appl Catal Gen 2019;570. https://doi.org/10.1016/j.apcata.2018.11.021.

[69] Li D, Shishido T, Oumi Y, Sano T, Takehira K. Self-activation and self-regenerative activity of trace Rh-doped Ni/Mg(Al)O catalysts in steam reforming of methane. Appl Catal Gen 2007;332. https://doi.org/10.1016/j.apcata.2007.08.008.

[70] Mozammel T, Dumbre D, Hubesch R, Yadav GD, Selvakannan PR, Bhargava SK. Carbon dioxide reforming of methane over mesoporous alumina supported Ni (Co), Ni(Rh) bimetallic, and Ni(CORh) trimetallic catalysts: role of nanoalloying in improving the stability and nature of coking. Energy Fuel 2020;34. https://doi.org/10.1021/acs.energyfuels.0c03249.

[71] Romano PN, de Carvalho Filho JFS, de Almeida JMAR, Sousa-Aguiar EF. Screening of mono and bimetallic catalysts for the dry reforming of methane. Catal Today 2022:394–6. https://doi.org/10.1016/j.cattod.2021.08.006.

[72] You X, Wang X, Ma Y, Liu J, Liu W, Xu X, et al. Ni-Co/Al2O3 bimetallic catalysts for CH4 steam reforming: elucidating the role of Co for improving coke resistance. ChemCatChem 2014;6. https://doi.org/10.1002/cctc.201402695.

[73] Wu Z, Yang B, Miao S, Liu W, Xie J, Lee S, et al. Lattice strained Ni-Co alloy as a high-performance catalyst for catalytic dry reforming of methane. ACS Catal 2019;9. https://doi.org/10.1021/acscatal.8b02821.

[74] Jiang C, Loisel E, Cullen DA, Dorman JA, Dooley KM. On the enhanced sulfur and coking tolerance of Ni-Co-rare earth oxide catalysts for the dry reforming of methane. J Catal 2021;393. https://doi.org/10.1016/j.jcat.2020.11.028.

[75] Beheshti Askari A, Al Samarai M, Morana B, Tillmann L, Pfänder N, Wandzilak A, et al. In situ X-ray microscopy reveals particle dynamics in a NiCo dry methane reforming catalyst under operating conditions. ACS Catal 2020;10. https://doi.org/10.1021/acscatal.9b05517.

[76] Beheshti Askari A, Al Samarai M, Hiraoka N, Ishii H, Tillmann L, Muhler M, et al. In situ X-ray emission and high-resolution X-ray absorption spectroscopy applied to Ni-based bimetallic dry methane reforming catalysts. Nanoscale 2020;12. https://doi.org/10.1039/d0nr01960g.

[77] Harshini D, Kwon Y, Han J, Yoon SP, Nam SW, Lim TH. Suppression of carbon formation in steam reforming of methane by addition of Co into Ni/ZrO2 catalysts. Kor J Chem Eng 2010;27. https://doi.org/10.2478/s11814-010-0095-9.

[78] Movasati A, Alavi SM, Mazloom G. Dry reforming of methane over CeO2-ZnAl2O4 supported Ni and Ni-Co nano-catalysts. Fuel 2019;236. https://doi.org/10.1016/j.fuel.2018.09.069.

[79] Wang H, Dong X, Zhao T, Yu H, Li M. Dry reforming of methane over bimetallic Ni-Co catalyst prepared from La(CoxNi1-x)0.5Fe0.5O3 perovskite precursor: catalytic activity and coking resistance. Appl Catal B Environ 2019;245. https://doi.org/10.1016/j.apcatb.2018.12.072.

[80] Das S, Sengupta M, Bag A, Shah M, Bordoloi A. Facile synthesis of highly disperse Ni-Co nanoparticles over mesoporous silica for enhanced methane dry reforming. Nanoscale 2018;10. https://doi.org/10.1039/c7nr09625a.

[81] Sheng K, Luan D, Jiang H, Zeng F, Wei B, Pang F, et al. NixCoy nanocatalyst supported by ZrO2 hollow sphere for dry reforming of methane: synergetic catalysis by Ni and Co in alloy. ACS Appl Mater Interfaces 2019;11. https://doi.org/10.1021/acsami.9b05822.

[82] Xu Y, Hirano T, Kunieda H, Hara Y, Miyata Y. Enhancing the methane steam reforming catalytic performance of Ni monolithic catalysts: via Ni-Re surface alloying. Catal Sci Technol 2020;10. https://doi.org/10.1039/c9cy02539a.

[83] Guo HH, Zhou D, Du C, Wang PJ, Liu WF, Pang LX, et al. Temperature stable Li2Ti0.75(Mg1/3Nb2/3)0.25O3-based microwave dielectric ceramics with low sintering temperature and ultra-low dielectric loss for dielectric resonator antenna applications. J Mater Chem C 2020;8. https://doi.org/10.1039/d0tc00326c.

[84] Zubenko D, Singh S, Rosen BA. Exsolution of Re-alloy catalysts with enhanced stability for methane dry reforming. Appl Catal B Environ 2017;209. https://doi.org/10.1016/j.apcatb.2017.03.047.

[85] Huang TJ, Yu TC, Jhao SY. Weighting variation of water-gas shift in steam reforming of methane over supported Ni and Ni-Cu catalysts. Ind Eng Chem Res 2006;45. https://doi.org/10.1021/ie050744r.

[86] Song K, Lu M, Xu S, Chen C, Zhan Y, Li D, et al. Effect of alloy composition on catalytic performance and coke-resistance property of Ni-Cu/Mg(Al)O catalysts for dry reforming of methane. Appl Catal B Environ 2018;239. https://doi.org/10.1016/j.apcatb.2018.08.023.

[87] Chatla A, Ghouri MM, El Hassan OW, Mohamed N, Prakash AV. Elbashir NO. An experimental and first principles DFT investigation on the effect of Cu addition to Ni/Al2O3 catalyst for the dry reforming of methane. Appl Catal Gen 2020;602. https://doi.org/10.1016/j.apcata.2020.117699.

[88] Rezaei R, Moradi G, Sharifnia S. Dry reforming of methane over Ni-Cu/Al2O3 catalyst coatings in a microchannel reactor: modeling and optimization using design of experiments. Energy Fuel 2019;33. https://doi.org/10.1021/acs.energyfuels.9b00692.

[89] Han K, Wang S, Liu Q, Wang F. Optimizing the Ni/Cu ratio in Ni-Cu nanoparticle catalysts for methane dry reforming. ACS Appl Nano Mater 2021;4. https://doi.org/10.1021/acsanm.1c00673.

[90] Wang M, Zhao T, Li M, Wang H. Perovskite La2(NiCu)O4 catalyst precursors for dry reforming of methane: effects of Cu-substitution on carbon resistance. RSC Adv 2017;7. https://doi.org/10.1039/c7ra08422f.

[91] Dam AH, Wang H, Dehghan-Niri R, Yu X, Walmsley JC, Holmen A, et al. Methane activation on bimetallic catalysts: properties and functions of surface Ni−Ag alloy. ChemCatChem 2019;11. https://doi.org/10.1002/cctc.201900679.

[92] Wang H, Blaylock DW, Dam AH, Liland SE, Rout KR, Zhu YA, et al. Steam methane reforming on a Ni-based bimetallic catalyst: density functional theory and experimental studies of the catalytic consequence of surface alloying of Ni with Ag. Catal Sci Technol 2017;7. https://doi.org/10.1039/c7cy00101k.

[93] Nikolla E, Schwank J, Linic S. Comparative study of the kinetics of methane steam reforming on supported Ni and Sn/Ni alloy catalysts: the impact of the formation of Ni alloy on chemistry. J Catal 2009;263. https://doi.org/10.1016/j.jcat.2009.02.006.

[94] Nikolla E, Schwank J, Linic S. Measuring and relating the electronic structures of nonmodel supported catalytic materials to their performance. J Am Chem Soc 2009;131. https://doi.org/10.1021/ja809291e.

[95] Guharoy U, Le Saché E, Cai Q, Reina TR, Gu S. Understanding the role of Ni-Sn interaction to design highly effective CO2 conversion catalysts for dry reforming of methane. J CO2 Util 2018;27. https://doi.org/10.1016/j.jcou.2018.06.024.

[96] da Silva F de AR, dos Santos RCR, Nunes RS, Valentini A. Role of tin on the electronic properties of Ni/Al2O3 catalyst and its effect over the methane dry reforming reaction. Appl Catal Gen 2021;618. https://doi.org/10.1016/j.apcata.2021.118129.

[97] Najfach AJ, Almquist CB, Edelmann RE. Effect of Manganese and zeolite composition on zeolite-supported Ni-catalysts for dry reforming of methane. Catal Today 2021;369. https://doi.org/10.1016/j.cattod.2020.07.058.

[98] Ramezani Y, Meshkani F, Rezaei M. Preparation and evaluation of mesoporous nickel and manganese bimetallic nanocatalysts in methane dry reforming process for syngas production. J Chem Sci 2018;130. https://doi.org/10.1007/s12039-017-1410-3.

[99] Jaiswar VK, Katheria S, Deo G, Kunzru D. Effect of Pt doping on activity and stability of Ni/MgAl2O4 catalyst for steam reforming of methane at ambient and high pressure condition. Int J Hydrogen Energy 2017;42. https://doi.org/10.1016/j.ijhydene.2017.06.096.

[100] Li X, Li D, Tian H, Zeng L, Zhao ZJ, Gong J. Dry reforming of methane over Ni/La2O3 nanorod catalysts with stabilized Ni nanoparticles. Appl Catal B Environ 2017;202. https://doi.org/10.1016/j.apcatb.2016.09.071.

[101] Roy S, Hariharan S, Tiwari AK. Pt-Ni subsurface alloy catalysts: an improved performance toward CH4 dissociation. J Phys Chem C 2018;122. https://doi.org/10.1021/acs.jpcc.8b01705.

[102] Egelske BT, Keels JM, Monnier JR, Regalbuto JR. An analysis of electroless deposition derived Ni-Pt catalysts for the dry reforming of methane. J Catal 2020;381. https://doi.org/10.1016/j.jcat.2019.10.035.

[103] Vasiliades MA, Damaskinos CM, Kyprianou KK, Kollia M, Efstathiou AM. The effect of Pt on the carbon pathways in the dry reforming of methane over Ni-Pt/Ce0.8Pr0.2O2-δ catalyst. Catal Today 2020;355. https://doi.org/10.1016/j.cattod.2019.04.022.

[104] Yao L, Galvez ME, Hu C, Da Costa P. Mo-promoted Ni/Al2O3 catalyst for dry reforming of methane. Int J Hydrogen Energy 2017;42. https://doi.org/10.1016/j.ijhydene.2017.03.208.

[105] Yao L, Galvez ME, Hu C, Da Costa P. Synthesis gas production via dry reforming of methane over manganese promoted nickel/cerium-zirconium oxide catalyst. Ind Eng Chem Res 2018;57. https://doi.org/10.1021/acs.iecr.8b04183.

[106] Song Y, Ozdemir E, Ramesh S, Adishev A, Subramanian S, Harale A, et al. Dry reforming of methane by stable Ni–Mo nanocatalysts on single-crystalline MgO. Science 2020;367. https://doi.org/10.1126/science.aav2412.

[107] de Oliveira Rocha K, Marques CMP, Bueno JMC. Effect of Au doping of Ni/Al2O3 catalysts used in steam reforming of methane: mechanism, apparent activation energy, and compensation effect. Chem Eng Sci 2019;207. https://doi.org/10.1016/j.ces.2019.06.049.

[108] Besenbacher F, Chorkendorff I, Clausen BS, Hammer B, Molenbroek AM, Nørskov JK, et al. Design of a surface alloy catalyst for steam reforming. Science 1998;279. https://doi.org/10.1126/science.279.5358.1913.

[109] Palma S, Bobadilla LF, Corrales A, Ivanova S, Romero-Sarria F, Centeno MA, et al. Effect of gold on a NiLaO3 perovskite catalyst for methane steam reforming. Appl Catal B Environ 2014;144. https://doi.org/10.1016/j.apcatb.2013.07.055.

[110] Chin YH, Cathy, King DL, Roh HS, Wang Y, Heald SM. Structure and reactivity investigations on supported bimetallic Au{single bond}Ni catalysts used for hydrocarbon steam reforming. J Catal 2006;244. https://doi.org/10.1016/j.jcat.2006.08.016.

[111] Sapountzi FM, Zhao C, Boréave A, Retailleau-Mevel L, Niakolas D, Neofytidis C, et al. Sulphur tolerance of Au-modified Ni/GDC during catalytic methane steam reforming. Catal Sci Technol 2018;8. https://doi.org/10.1039/c8cy00107c.

[112] Wang ZT, Darby MT, Therrien AJ, El-Soda M, Michaelides A, Stamatakis M, et al. Preparation, structure, and surface chemistry of Ni-Au single atom alloys. J Phys Chem C 2016;120. https://doi.org/10.1021/acs.jpcc.6b03473.

[113] Liu W, Li L, Lin S, Luo Y, Bao Z, Mao Y, et al. Confined Ni-In intermetallic alloy nanocatalyst with excellent coking resistance for methane dry reforming. J Energy Chem 2021;65. https://doi.org/10.1016/j.jechem.2021.05.017.

[114] Károlyi J, Németh M, Evangelisti C, Sáfrán G, Schay Z, Horváth A, et al. Carbon dioxide reforming of methane over Ni–In/SiO2 catalyst without coke formation. J Ind Eng Chem 2018;58. https://doi.org/10.1016/j.jiec.2017.09.024.

[115] Sokolov S, Radnik J, Schneider M, Rodemerck U. Low-temperature CO2 reforming of methane over Ni supported on ZnAl mixed metal oxides. Int J Hydrogen Energy 2017;42. https://doi.org/10.1016/j.ijhydene.2017.01.013.

[116] Masoom Nataj SM, Alavi SM, Mazloom G. Catalytic performance of Ni supported on ZnO-Al 2 O 3 composites with different Zn content in methane dry reforming. J Chem Technol Biotechnol 2019;94. https://doi.org/10.1002/jctb.5887.

[117] Tada M, Zhang S, Malwadkar S, Ishiguro N, Soga JI, Nagai Y, et al. The active phase of nickel/ordered Ce 2Zr 2O x catalysts with a discontinuity (x=7 - 8) in

methane steam reforming. Angew Chem Int Ed 2012;51. https://doi.org/10.1002/anie.201205167.
[118] Sutthiumporn K, Maneerung T, Kathiraser Y, Kawi S. CO2 dry-reforming of methane over La0.8Sr 0.2Ni0.8M0.2O3 perovskite (M = Bi, Co, Cr, Cu, Fe): roles of lattice oxygen on C-H activation and carbon suppression. Int J Hydrogen Energy 2012;37. https://doi.org/10.1016/j.ijhydene.2012.04.059.
[119] Lazar MD, Dan M, Mihet M, Almasan V, Rednic V, Borodi G. Hydrogen production by low temperature methane steam reforming using Ag and Au modified alumina supported nickel catalysts. Rev Roum Chem 2011;56.
[120] Lazar M, Mihet M, Dan M, Almasan V, Marginean P. Preparation and characterization of nickel based multicomponent catalysts. J Phys Conf 2009;182. https://doi.org/10.1088/1742-6596/182/1/012049.
[121] Dan M, Mihet M, Biris AR, Marginean P, Almasan V, Borodi G, et al. Supported nickel catalysts for low temperature methane steam reforming: comparison between metal additives and support modification. React Kinet Mech Catal 2012; 105. https://doi.org/10.1007/s11144-011-0406-0.
[122] Parizotto NV, Rocha KO, Damyanova S, Passos FB, Zanchet D, Marques CMP, et al. Alumina-supported Ni catalysts modified with silver for the steam reforming of methane: effect of Ag on the control of coke formation. Appl Catal Gen 2007; 330. https://doi.org/10.1016/j.apcata.2007.06.022.
[123] Maniecki TP, Stadnichenko AI, Maniukiewicz W, Bawolak K, Mierczyński P, Boronin AI, et al. An active phase transformation on surface of Ni-Au/Al2O 3 catalyst during partial oxidation of methane to synthesis gas. Kinet Catal 2010;51. https://doi.org/10.1134/S0023158410040178.
[124] Wang W, Su C, Ran R, Shao Z. A new Gd-promoted nickel catalyst for methane conversion to syngas and as an anode functional layer in a solid oxide fuel cell. J Power Sources 2011;196. https://doi.org/10.1016/j.jpowsour.2010.12.072.
[125] Kim HW, Kang KM, Kwak HY, Kim JH. Preparation of supported Ni catalysts on various metal oxides with core/shell structures and their tests for the steam reforming of methane. Chem Eng J 2011;168. https://doi.org/10.1016/j.cej.2010.11.045.
[126] Roh HS, Jun KW, Park SE. Methane-reforming reactions over Ni/Ce-ZrO2/θ-Al 2O3 catalysts. Appl Catal Gen 2003;251. https://doi.org/10.1016/S0926-860X(03)00359-4.
[127] Zhai X, Ding S, Liu Z, Jin Y, Cheng Y. Catalytic performance of Ni catalysts for steam reforming of methane at high space velocity. Int J Hydrogen Energy 2011; 36. https://doi.org/10.1016/j.ijhydene.2010.10.053.
[128] Bej B, Pradhan NC, Neogi S. Production of hydrogen by steam reforming of methane over alumina supported nano-NiO/SiO2 catalyst. Catal Today 2013;207. https://doi.org/10.1016/j.cattod.2012.04.011.
[129] Salhi N, Boulahouache A, Petit C, Kiennemann A, Rabia C. Steam reforming of methane to syngas over NiAl2O4 spinel catalysts. Int J Hydrogen Energy 2011;36. https://doi.org/10.1016/j.ijhydene.2010.11.071.
[130] Ali S, Al-Marri MJ, Abdelmoneim AG, Kumar A, Khader MM. Catalytic evaluation of nickel nanoparticles in methane steam reforming. Int J Hydrogen Energy 2016; 41:22876–85. https://doi.org/10.1016/j.ijhydene.2016.08.200.
[131] Nazari M, Alavi SM. An investigation of the simultaneous presence of Cu and Zn in different Ni/Al2O3 catalyst loads using Taguchi design of experiment in steam reforming of methane. Int J Hydrogen Energy 2020;45. https://doi.org/10.1016/j.ijhydene.2019.10.224.
[132] Katheria S, Gupta A, Deo G, Kunzru D. Effect of calcination temperature on stability and activity of Ni/MgAl2O4 catalyst for steam reforming of methane at high pressure condition. Int J Hydrogen Energy 2016;41:14123–32. https://doi.org/10.1016/j.ijhydene.2016.05.109.
[133] Boudjeloud M, Boulahouache A, Rabia C, Salhi N. La-doped supported Ni catalysts for steam reforming of methane. Int J Hydrogen Energy 2019;44. https://doi.org/10.1016/j.ijhydene.2019.01.140.
[134] Rogers JL, Mangarella MC, D'Amico AD, Gallagher JR, Dutzer MR, Stavitski E, et al. Differences in the nature of active sites for methane dry reforming and methane steam reforming over nickel aluminate catalysts. ACS Catal 2016;6. https://doi.org/10.1021/acscatal.6b01133.
[135] Azancot L, Bobadilla LF, Santos JL, Córdoba JM, Centeno MA, Odriozola JA. Influence of the preparation method in the metal-support interaction and reducibility of Ni-Mg-Al based catalysts for methane steam reforming. Int J Hydrogen Energy 2019;44. https://doi.org/10.1016/j.ijhydene.2019.05.167.
[136] Fang X, Zhang X, Guo Y, Chen M, Liu W, Xu X, et al. Highly active and stable Ni/Y2Zr2O7 catalysts for methane steam reforming: on the nature and effective preparation method of the pyrochlore support. Int J Hydrogen Energy 2016;41. https://doi.org/10.1016/j.ijhydene.2016.04.038.
[137] Lertwittayanon K, Youravong W, Lau WJ. Enhanced catalytic performance of Ni/A-Al2O3 catalyst modified with CaZrO3 nanoparticles in steam-methane reforming. Int J Hydrogen Energy 2017;42. https://doi.org/10.1016/j.ijhydene.2017.09.030.
[138] Iglesias I, Baronetti G, Mariño F. Ni/Ce0.95M0.05O2−d (M = Zr, Pr, La) for methane steam reforming at mild conditions. Int J Hydrogen Energy 2017;42. https://doi.org/10.1016/j.ijhydene.2017.09.176.
[139] Yoo J, Park S, Song JH, Yoo S, Song IK. Hydrogen production by steam reforming of natural gas over butyric acid-assisted nickel/alumina catalyst. Int J Hydrogen Energy 2017;42. https://doi.org/10.1016/j.ijhydene.2017.09.148.
[140] Li W, Zhao Z, Ding F, Guo X, Wang G. Syngas production via steam-CO2 dual reforming of methane over LA-Ni/ZrO2 catalyst prepared by l -arginine ligand-assisted strategy: enhanced activity and stability. ACS Sustainable Chem Eng 2015;3. https://doi.org/10.1021/acssuschemeng.5b01277.
[141] Park YS, Kang M, Byeon P, Chung SY, Nakayama T, Ko T, et al. Fabrication of a regenerable Ni supported NiO-MgO catalyst for methane steam reforming by exsolution. J Power Sources 2018;397. https://doi.org/10.1016/j.jpowsour.2018.07.025.
[142] Nurmakanov YY, Yemelyanova VS, Itkulova SS, Zakumbaeva GD, Nurgaliyev NN. Carbon dioxide conversion of biogas with producing syngas over the polymetallic supported catalysts. https://doi.org/10.1007/978-3-319-16901-9_61; 2015.
[143] Zhang X, Peng L, Fang X, Cheng Q, Liu W, Peng H, et al. Ni/Y2B2O7 (B[dbnd]Ti, Sn, Zr and Ce) catalysts for methane steam reforming: on the effects of B site replacement. Int J Hydrogen Energy 2018;43. https://doi.org/10.1016/j.ijhydene.2018.03.086.
[144] Al-Najar AMA, Al-Doghachi FAJ, Al-Riyahee AAA, Taufiq-Yap YH. Effect of La2O3 as a promoter on the Pt,Pd,Ni/MgO catalyst in dry reforming of methane reaction. Catalysts 2020;10. https://doi.org/10.3390/catal10070750.
[145] Chen C, Wang X, Chen X, Liang X, Zou X, Lu X. Combined steam and CO2 reforming of methane over one-pot prepared Ni/La-Si catalysts. Int J Hydrogen Energy 2019;44. https://doi.org/10.1016/j.ijhydene.2019.01.019.
[146] Sebai I, Boulahaouache A, Trari M, Salhi N. Preparation and characterization of 5%Ni/Γ-Al2o3 catalysts by complexation with NH3 derivatives active in methane steam reforming. Int J Hydrogen Energy 2019;44. https://doi.org/10.1016/j.ijhydene.2018.12.050.
[147] Fang X, Xu L, Zhang X, Zhang K, Dai H, Liu W, et al. Effect of rare earth element (Ln = La, Pr, Sm, and Y) on physicochemical properties of the Ni/Ln2Ti2O7 catalysts for the steam reforming of methane. Mol Catal 2019;468. https://doi.org/10.1016/j.mcat.2019.02.022.
[148] Noh YS, Lee KY, Moon DJ. Hydrogen production by steam reforming of methane over nickel based structured catalysts supported on calcium aluminate modified SiC. Int J Hydrogen Energy 2019;44. https://doi.org/10.1016/j.ijhydene.2019.04.287.
[149] Dan M, Mihet M, Lazar MD. Hydrogen and/or syngas production by combined steam and dry reforming of methane on nickel catalysts. Int J Hydrogen Energy 2020;45. https://doi.org/10.1016/j.ijhydene.2019.12.158.
[150] Khani Y, Shariatinia Z, Bahadoran F. High catalytic activity and stability of ZnLaAlO4 supported Ni, Pt and Ru nanocatalysts applied in the dry, steam and combined dry-steam reforming of methane. Chem Eng J 2016;299. https://doi.org/10.1016/j.cej.2016.04.108.
[151] Meloni E, Martino M, Palma V. A short review on ni based catalysts and related engineering issues for methane steam reforming. Catalysts 2020;10. https://doi.org/10.3390/catal10030352.
[152] Tang X, Wu Y, Fang Z, Dong X, Du Z, Deng B, et al. Syntheses, catalytic performances and DFT investigations: a recent review of copper-based catalysts of methanol steam reforming for hydrogen production. Energy 2024;295:131091. https://doi.org/10.1016/j.energy.2024.131091.
[153] Pashchenko D. Intra-particle diffusion limitation for steam methane reforming over a Ni-based catalyst. Fuel 2023;353. https://doi.org/10.1016/j.fuel.2023.129205.
[154] Farinha JF. Methane steam reforming in laboratory scale n.d:1–10..
[155] Costamagna P, Pugliese F, Cavattoni T, Busca G, Garbarino G. Modeling of laboratory steam methane reforming and CO2 methanation reactors. Energies 2020;13. https://doi.org/10.3390/en13102624.
[156] Ściążko A, Komatsu Y, Brus G, Kimijima S, Szmyd J. Analiza procesu reformingu metanu parą wodną przy pomocy uogólnionej metody najmniejszych kwadratów. Scientific Letters of Rzeszow University of Technology - Mechanics 2014;31: 95–102. https://doi.org/10.7862/rm.2014.11.
[157] Park HG, Han SY, Jun KW, Woo Y, Park MJ, Kim SK. Bench-scale steam reforming of methane for hydrogen production. Catalysts 2019;9. https://doi.org/10.3390/catal9070615.
[158] Garbarino G, Pugliese F, Cavattoni T, Busca G, Costamagna P. A study on CO2 methanation and steam methane reforming over commercial Ni/calcium aluminate catalysts. Energies 2020;13. https://doi.org/10.3390/en13112792.
[159] Belhadi A, Boumaza S, Djadoun A, Trari M, Cherifi O. Methane steam reforming on supported nickel, effect of nickel content for product hydrogen. Open J Phys Chem 2016. https://doi.org/10.4236/ojpc.2016.62003. 06.
[160] Dong WS, Roh HS, Jun KW, Park SE, Oh YS. Methane reforming over Ni/Ce-ZrO2 catalysts: effect of nickel content. Appl Catal Gen 2002;226. https://doi.org/10.1016/S0926-860X(01)00883-3.
[161] Lemonidou AA, Vasalos IA. Carbon dioxide reforming of methane over 5 wt.% Ni/CaO-Al2O3 catalyst. Appl Catal Gen 2002;228. https://doi.org/10.1016/S0926-860X(01)00974-7.
[162] Oliveira ELG, Grande CA, Rodrigues AE. Methane steam reforming in large pore catalyst. Chem Eng Sci 2010;65. https://doi.org/10.1016/j.ces.2009.10.018.
[163] Castro Luna AE, Becerra AM. Kinetics of methane steam reforming on a Ni on alumina-titania catalyst. React Kinet Catal Lett 1997;61. https://doi.org/10.1007/BF02478395.
[164] Szablowski L, Dybinski O, Szczesniak A, Milewski J. Mathematical model of steam reforming in the anode channel of a molten carbonate fuel cell. Energies 2022;15. https://doi.org/10.3390/en15020608.
[165] Abbas SZ, Dupont V, Mahmud T. Kinetics study and modelling of steam methane reforming process over a NiO/Al2O3 catalyst in an adiabatic packed bed reactor. Int J Hydrogen Energy 2017;42. https://doi.org/10.1016/j.ijhydene.2016.11.093.
[166] Szablowski L, Kupecki J, Milewski J, Motylinski K. Kinetic model of a plate fin heat exchanger with catalytic coating as a steam reformer of methane, biogas, and dimethyl ether. Int J Energy Res 2019;43. https://doi.org/10.1002/er.4465.
[167] Pashchenko D. Combined methane reforming with a mixture of methane combustion products and steam over a Ni-based catalyst: an experimental and thermodynamic study. Energy 2019;185. https://doi.org/10.1016/j.energy.2019.07.065.

[168] Pashchenko D. Experimental investigation of synthesis gas production by methane reforming with flue gas over a NiO-Al 2 O 3 catalyst: reforming characteristics and pressure drop. Int J Hydrogen Energy 2019;44. https://doi.org/10.1016/j.ijhydene.2019.01.250.
[169] Wismann ST, Engbæk JS, Vendelbo SB, Eriksen WL, Frandsen C, Mortensen PM, et al. Electrified methane reforming: elucidating transient phenomena. Chem Eng J 2021;425. https://doi.org/10.1016/j.cej.2021.131509.
[170] Zheng L, Ambrosetti M, Marangoni D, Beretta A, Groppi G, Tronconi E. Electrified methane steam reforming on a washcoated SiSiC foam for low-carbon hydrogen production. AIChE J 2023;69. https://doi.org/10.1002/aic.17620.
[171] IEA. Global hydrogen review 2021 - International Energy Agency. n.d..
[172] Cox. Hydrogen: its technology and implications. https://doi.org/10.1201/9781351073301; 2018.
[173] Shell launches quest carbon capture and storage project. Online n.d.
[174] Milewski J, Szczęśniak A, Martsinchyk A, Bartela Ł, Zdeb J, Dybiński O. Pilot-scale SOE-MCFC hybrid system for Co2/H2 mixture production – first experiences in the "Tennessee" project. Int J Hydrogen Energy 2024;52. https://doi.org/10.1016/j.ijhydene.2023.10.054.
[175] Milewski J, Zdeb J, Szczęśniak A, Martsinchyk A, J Kupecki OD. Concept of a solid oxide electrolysis-molten carbonate fuel cell hybrid system to support a power-to-gas installation. Energy Convers Manag 2023;276.
[176] Larminie J, Dicks A. Fuel cell systems explained. second ed. 2013. https://doi.org/10.1002/9781118878330.
[177] Rostrup-Nielsen JR. Catalytic steam reforming. Catal Sci Technol 1984. https://doi.org/10.1007/978-3-642-93247-2_1.
[178] Luo M, Yi Y, Wang S, Wang Z, Du M, Pan J, et al. Review of hydrogen production using chemical-looping technology. Renew Sustain Energy Rev 2018;81. https://doi.org/10.1016/j.rser.2017.07.007.
[179] Łukasiewicz Research Network – New Chemical Syntheses Institute n.d. https://ins.lukasiewicz.gov.pl/index.php/en/products/catalysts-and-sorbents/hydrocarbon-steam-reforming.
[180] Clariant Catalysts (www.clariant.com) n.d.
[181] Umicore (www.fcs.umicore.com) n.d.
[182] Johnson Matthey (www.matthey.com) n.d.
[183] Tanaka Precious Metals (www.tanaka-preciousmetals.com) n.d.
[184] Mozdzierz M, Brus G, Sciazko A, Komatsu Y, Kimijima S, Szmyd JS. Towards a thermal optimization of a methane/steam reforming reactor. Flow, Turbul Combust 2016;97. https://doi.org/10.1007/s10494-015-9693-2.
[185] Izquierdo U, Barrio VL, Cambra JF, Requies J, Güemez MB, Arias PL, et al. Hydrogen production from methane and natural gas steam reforming in conventional and microreactor reaction systems. Int J Hydrogen Energy 2012;37. https://doi.org/10.1016/j.ijhydene.2011.11.048.
[186] Pashchenko D. Flow dynamic in a packed bed filled with Ni-Al 2 O 3 porous catalyst: experimental and numerical approach. AIChE J 2019;65. https://doi.org/10.1002/aic.16558.
[187] Pashchenko D, Karpilov I, Mustafin R. Numerical calculation with experimental validation of pressure drop in a fixed-bed reactor filled with the porous elements. AIChE J 2020;66. https://doi.org/10.1002/aic.16937.
[188] Rosen MA. Thermodynamic investigation of hydrogen production by steam-methane reforming. Int J Hydrogen Energy 1991;16. https://doi.org/10.1016/0360-3199(91)90003-2.
[189] Spath PL, Mann MK. Life cycle assessment of hydrogen production via natural gas steam reforming. https://doi.org/10.2172/764485; 2000.
[190] Pashchenko D, Nikitin M. Forging furnace with thermochemical waste-heat recuperation by natural gas reforming: fuel saving and heat balance. Int J Hydrogen Energy 2021;46. https://doi.org/10.1016/j.ijhydene.2020.09.228.
[191] Sun Y, Zhang Z, Liu L, Wang X. Heat recovery from high temperature slags: a review of chemical methods. Energies 2015;8. https://doi.org/10.3390/en8031917.
[192] Zhang B, Lu Q, Huang W, Chen Y, Yan W, Yu B, et al. Hydrogen production via in-line pyrolysis-reforming of organic solid waste enhanced by steel slags. Int J Hydrogen Energy 2022;47. https://doi.org/10.1016/j.ijhydene.2021.12.065.
[193] Zhang B, Chen Y, Zhang B, Peng R, Lu Q, Yan W, et al. Cyclic performance of coke oven gas - steam reforming with assistance of steel slag derivates for high purity hydrogen production. Renew Energy 2022;184. https://doi.org/10.1016/j.renene.2021.11.123.
[194] Zhang B, Chen Y, Kang B, Qian J, Chuai X, Peng R, et al. Hydrogen production via steam reforming of coke oven gas enhanced by steel slag-derived CaO. Int J Hydrogen Energy 2020;45. https://doi.org/10.1016/j.ijhydene.2020.03.061.
[195] Pashchenko D. A heat recovery rate of the thermochemical waste-heat recuperation systems based on experimental prediction. Energy 2020;198. https://doi.org/10.1016/j.energy.2020.117395.
[196] Tang Y, Wang Y, Long W, Xiao G, Wang Y, Li W. Analysis and enhancement of methanol reformer performance for online reforming based on waste heat recovery of methanol-diesel dual direct injection engine. Energy 2023;283. https://doi.org/10.1016/j.energy.2023.129098.
[197] Gallucci F, van Sint Annaland M, Roses L, Manzolini G. Using palladium membrane-based fuel reformers for combined heat and power (CHP) plants. Palladium Membrane Technology for Hydrogen Production, Carbon Capture and Other Applications: Principles, Energy Production and Other Applications 2015. https://doi.org/10.1533/9781782422419.2.319.
[198] Fan R, Li Z-Q, Zhang H-M, Tao W-Q. Analysis of a combined heating and power system based on high-temperature proton exchange membrane fuel cell and steam methane reforming: from energy, exergy and economic point of views. Appl Therm Eng 2024;247:123075. https://doi.org/10.1016/j.applthermaleng.2024.123075.
[199] Wang Y, Su B, Yang X, Chen Z, Wang S. A green hydrogen production system based on solar-assisted biogas steam reforming. Appl Therm Eng 2024;248:123067. https://doi.org/10.1016/j.applthermaleng.2024.123067. Part.
[200] Natrella G, Borgogna A, Salladini A, Iaquaniello G. How to give a renewed chance to natural gas as feed for the production of hydrogen: electric MSR coupled with CO2 mineralization. Cleaner Engineering and Technology 2021;5. https://doi.org/10.1016/j.clet.2021.100280.
[201] Zhang Y, Zhang S, Lou HH, Gossage JL, Benson TJ. Steam and dry reforming processes coupled with partial oxidation of methane for CO2 emission reduction. Chem Eng Technol 2014;37. https://doi.org/10.1002/ceat.201400132.
[202] Roh HS, Lee DK, Koo KY, Jung UH, Yoon WL. Natural gas steam reforming for hydrogen production over metal monolith catalyst with efficient heat-transfer. Int J Hydrogen Energy 2010;35. https://doi.org/10.1016/j.ijhydene.2009.12.051.
[203] Rajesh JK, Gupta SK, Rangaiah GP, Ray AK. Multi-objective optimization of industrial hydrogen plants. Chem Eng Sci 2001;56. https://doi.org/10.1016/S0009-2509(00)00316-X.
[204] Roh HS, Jun KW. Low temperature methane steam reforming for hydrogen production for fuel cells. Bull Kor Chem Soc 2009;30. https://doi.org/10.5012/bkcs.2009.30.1.153.
[205] Collodi G. Hydrogen production via steam reforming with CO2 capture. Chemical Engineering Transactions 2010;19. https://doi.org/10.3303/CET1019007.
[206] Yan Y, Thanganadar D, Clough PT, Mukherjee S, Patchigolla K, Manovic V, et al. Process simulations of blue hydrogen production by upgraded sorption enhanced steam methane reforming (SE-SMR) processes. Energy Convers Manag 2020;222. https://doi.org/10.1016/j.enconman.2020.113144.
[207] Song C, Liu Q, Ji N, Kansha Y, Tsutsumi A. Optimization of steam methane reforming coupled with pressure swing adsorption hydrogen production process by heat integration. Appl Energy 2015;154. https://doi.org/10.1016/j.apenergy.2015.05.038.
[208] Subraveti SG, Roussanaly S, Anantharaman R, Riboldi L, Rajendran A. Techno-economic assessment of optimised vacuum swing adsorption for post-combustion CO2 capture from steam-methane reformer flue gas. Separ Purif Technol 2021;256. https://doi.org/10.1016/j.seppur.2020.117832.
[209] Chen Y, Zhao Y, Zhang J, Zheng C. Hydrogen production through CO2 sorption-enhanced methane steam reforming: comparison between different adsorbents. Sci China Technol Sci 2011;54. https://doi.org/10.1007/s11431-011-4587-6.
[210] Romano MC, Chiesa P, Lozza G. Pre-combustion CO2 capture from natural gas power plants, with ATR and MDEA processes. Int J Greenh Gas Control 2010;4. https://doi.org/10.1016/j.ijggc.2010.04.015.
[211] Nwaoha C, Tontiwachwuthikul P, Benamor A. CO2 capture from water-gas shift process plant: comparative bench-scale pilot plant investigation of MDEA-PZ blend vs novel MDEA activated by 1,5-diamino-2-methylpentane. Int J Greenh Gas Control 2019;82. https://doi.org/10.1016/j.ijggc.2019.01.009.
[212] Antonini C, Pérez-Calvo JF, van der Spek M, Mazzotti M. Optimal design of an MDEA CO2 capture plant for low-carbon hydrogen production — a rigorous process optimization approach. Separ Purif Technol 2021;279. https://doi.org/10.1016/j.seppur.2021.119715.
[213] Benamor A, Nasser M, Al-Marri MJ. Gas processing technology-treatment and utilization. Encyclopedia of Sustainable Technologies 2017. https://doi.org/10.1016/B978-0-12-409548-9.10101-0.
[214] Nord LO, Kothandaraman A, Herzog H, McRae G, Bolland O. A modeling software linking approach for the analysis of an integrated reforming combined cycle with hot potassium carbonate CO2 capture. Energy Proc 2009;1. https://doi.org/10.1016/j.egypro.2009.01.098.
[215] Vega F, Cano M, Camino S, Fernández LMG, Portillo E, Navarrete B. Solvents for carbon dioxide capture. Carbon Dioxide Chemistry, Capture and Oil Recovery 2018. https://doi.org/10.5772/intechopen.71443.
[216] Johannessen E, Jordal K. Study of a H2 separating membrane reactor for methane steam reforming at conditions relevant for power processes with CO2 capture. Energy Convers Manag 2005;46. https://doi.org/10.1016/j.enconman.2004.06.030.
[217] Najmi B, Soltanieh M. Process integration of membrane reactor for steam methane reforming for hydrogen separation with CO2 capture in power production by natural gas combined cycle. Energy Proc 2009;1. https://doi.org/10.1016/j.egypro.2009.01.039.
[218] Wang H, Hao Y. Thermodynamic study of solar thermochemical methane steam reforming with alternating H2 and CO2 permeation membranes reactors. Energy Proc 2017;105. https://doi.org/10.1016/j.egypro.2017.03.570.
[219] Ovalle-Encinia O, Wu HC, Chen T, Lin JYS. CO2-permselective membrane reactor for steam reforming of methane. J Membr Sci 2022;641. https://doi.org/10.1016/j.memsci.2021.119914.
[220] Do HY, Kim CH, Han JY, Kim HS, Ryi SK. Low-temperature proton-exchange membrane fuel cell-grade hydrogen production by membrane reformer equipped with Pd-composite membrane and methanation catalyst on permeation stream. J Membr Sci 2021;634. https://doi.org/10.1016/j.memsci.2021.119373.
[221] Cherif A, Atwair M, Atsbha TA, Zarei M, Duncan IJ, Nebbali R, et al. Enabling low-carbon membrane steam methane reforming: comparative analysis and multi-objective NSGA-II-integrated Bayesian optimization. Energy Convers Manag 2023;297. https://doi.org/10.1016/j.enconman.2023.117718.
[222] Wei Q, Yang G, Gao X, Yamane N, Zhang P, Liu G, et al. Ni/Silicalite-1 coating being coated on SiC foam: a tailor-made monolith catalyst for syngas production using a combined methane reforming process. Chem Eng J 2017;327. https://doi.org/10.1016/j.cej.2017.06.109.
[223] Chang JS, Park SE, Chon H. Catalytic activity and coke resistance in the carbon dioxide reforming of methane to synthesis gas over zeolite-supported Ni catalysts. Appl Catal Gen 1996;145. https://doi.org/10.1016/0926-860X(96)00150-0.

[224] González AR, Asencios YJO, Assaf EM, Assaf JM. Dry reforming of methane on Ni-Mg-Al nano-spheroid oxide catalysts prepared by the sol-gel method from hydrotalcite-like precursors. Appl Surf Sci 2013;280. https://doi.org/10.1016/j.apsusc.2013.05.082.

[225] Dębek R, Motak M, Galvez ME, Da Costa P, Grzybek T. Catalytic activity of hydrotalcite-derived catalysts in the dry reforming of methane: on the effect of Ce promotion and feed gas composition. React Kinet Mech Catal 2017;121. https://doi.org/10.1007/s11144-017-1167-1.

[226] Dębek R, Galvez ME, Launay F, Motak M, Grzybek T, Da Costa P. Low temperature dry methane reforming over Ce, Zr and CeZr promoted Ni–Mg–Al hydrotalcite-derived catalysts. Int J Hydrogen Energy 2016;41. https://doi.org/10.1016/j.ijhydene.2016.02.074.

[227] De Souza VP, Costa D, Dos Santos D, Sato AG, Bueno JMC. Pt-promoted α-Al 2O 3-supported Ni catalysts: effect of preparation conditions on oxi-reduction and catalytic properties for hydrogen production by steam reforming of methane. Int J Hydrogen Energy 2012;37. https://doi.org/10.1016/j.ijhydene.2012.03.141.

[228] Bengaard HS, Nørskov JK, Sehested J, Clausen BS, Nielsen LP, Molenbroek AM, et al. Steam reforming and graphite formation on Ni catalysts. J Catal 2002;209. https://doi.org/10.1006/jcat.2002.3579.